Manifold-Aware Concept Erasure

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Abstract

Concept erasure aims to remove a target concept from a representation with minimal damage to the rest. Unconstrained ambient-space updates can distort unrelated concepts; we propose TanGCE, a tangent-constrained method that restricts every erasure step to a local tangent approximation estimated by SVD of nearest-neighbor secants and reweights it by the local singular spectrum, so each axis is updated proportional to the manifold's local extent. A two-stage closed-form preprocessing (LEACE + rank-2 covariance-asymmetry projection) exhausts 1st- and 2nd-moment concept signal before the iterative tangent loop runs. We evaluate on 119 panels: 13 language models (0.5B–27B parameters) $\times$ 3 concepts (sycophancy, gender, safety) at 50%-depth, plus 40 binary CelebA-CLIP attributes $\times$ 2 control-set regimes — all under the same fixed pipeline hyperparameters with no per-panel tuning. TanGCE reaches the majority-class probe-accuracy floor at every control-damage budget on 26 to 33 of 39 NLP panels (vs. 22 to 26 for the strongest published unsupervised baseline) while damaging the control task less, and is the only method that retains an admissible operating point on every CelebA attribute when target and controls are highly correlated.

1 Introduction

Neural representations encode many attributes of their inputs: semantic content, style, demographic information, task labels, and other latent factors. In many applications, we may want to remove one such attribute, for example a protected or spurious concept, while preserving everything else the representation contains [De-Arteaga et al., 2019, Avitan et al., 2025]. This is the central tension in concept erasure. The goal is not merely to make one trained probe fail, but to remove the target information in a way that does not unnecessarily damage unknown control concepts.

This requirement makes erasure especially difficult. A target concept is rarely encoded in isolation. It may be entangled with many other factors, most of which are unlabeled and therefore cannot be explicitly protected. Moreover, erasure must address *recoverability*: even when a chosen classifier is driven to chance, a freshly trained classifier on the edited representations may still recover the protected attribute [Elazar and Goldberg, 2018]. Linear erasure methods provide simple and effective closed-form interventions [Ravfogel et al., 2020, Belrose et al., 2023], but they only remove linearly accessible information and do not generally block nonlinear probes [Ravfogel et al., 2022]. Nonlinear erasure, however, introduces a different danger: if we directly follow the gradient of a nonlinear concept scorer in the ambient representation space, the edit may remove target signal by moving the representation to a region that natural inputs never occupy.

Our starting point is that hidden states live in $\mathbb{R}^d$, but natural representations do not fill this space uniformly. Instead, they concentrate near a much smaller, structured support: the region of representation space that is actually reachable from natural inputs. This observation changes the geometry of erasure. An arbitrary direction in the ambient space may be a valid vector in $\mathbb{R}^d$, but it need not be a valid direction along the representation support. Moving in such a direction can push the representation off-support, distorting unrelated information and harming control concepts. We therefore ask whether

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nonlinear erasure can be made more surgical by forcing the edit to remain locally compatible with the geometry of natural representations.

We introduce TanGCE, a tangent gradient-based method for nonlinear concept erasure. Around each representation, we estimate a local tangent space using the SVD of nearest-neighbor secants. Instead of applying the ambient gradient of a concept scorer, we project that gradient onto the estimated tangent space and edit only along directions supported by the local neighborhood. This constrains erasure to directions that are locally plausible under the representation manifold, reducing the risk of off-support interventions.

A second ingredient is needed for stability. The local tangent space is not isotropic: different tangent directions correspond to different local scales of variation. A step of the same Euclidean length can be harmless along a high-variance direction but destructive along a low-variance one. We therefore use the singular spectrum produced by the same local SVD to precondition the tangent step. After projecting the scorer gradient onto the tangent space, we express it in the local principal-coordinate system and rescale each coordinate by its corresponding local scale. This local-σ preconditioner ties the magnitude of the intervention in each direction to how far the representation neighborhood actually extends in that direction, yielding a per-row trust region that transfers across models and layers.

Our full pipeline has two stages. First, we remove dominant low-order target signal in closed form. We apply LEACE to remove the first-order mean-difference direction between the two target classes [Belrose et al., 2023], and then remove a rank-2 subspace corresponding to the leading covariance-asymmetry directions of the class-conditional covariance difference. This sacrifices only a small number of representation dimensions while reducing the strongest linear and quadratic traces of the target concept. Second, we run tangent-constrained nonlinear erasure on the remaining signal using the local-σ preconditioned tangent gradients described above.

The empirical signature of this design is surgicality. We evaluate TanGCE on 119 panels under fixed pipeline hyperparameters, without per-panel tuning: 13 language models $\times$ 3 concepts at 50%-depth, and CelebA-CLIP $\times$ 40 attributes $\times$ 2 control-set regimes. Across control-damage budgets, TanGCE reaches the per-panel majority-class probe-accuracy floor on more panels than the strongest published unsupervised baseline while preserving control-task accuracy. On the NLP panels, TanGCE reaches the floor on 26–33/39 panels, compared with 22–26/39 for Obliviator, with stronger control preservation.

Our contributions are:

1. We formulate nonlinear concept erasure as a geometry-constrained intervention problem: target information should be removed without moving representations away from the local support of natural inputs.
2. We introduce TanGCE, a two-stage erasure pipeline combining closed-form low-order preprocessing with tangent-constrained nonlinear erasure.
3. We propose a local-σ preconditioner that scales tangent-coordinate steps by the local singular spectrum, making the trust region sensitive to the manifold's anisotropic geometry.
4. We evaluate TanGCE on 119 panels under a single fixed-ε protocol, showing stronger target removal and control preservation than Obliviator across NLP and vision-language settings.

2 Related Work

Concept erasure methods differ mainly in the function class against which they erase and in the geometry they assume. Linear methods such as INLP [Ravfogel et al., 2020] and LEACE [Belrose et al., 2023] target linearly decodable concept information. INLP removes successively exposed linear directions by iterative nullspace projection, while LEACE gives the strongest closed-form linear baseline in our comparisons. These methods are essential reference points, but they do not directly address the case where a fresh nonlinear probe can still recover the concept after linear erasure.

That recoverability concern is central to our framing. Elazar and Goldberg [Elazar and Goldberg, 2018] showed that apparent success against one learned remover or classifier does not imply that the protected information is gone for a newly trained downstream model. Amnesic probing [Elazar et al., 2021] and related intervention-based analyses adopt the same conservative lesson: the meaningful

Algorithm 1 TanGCE at a glance. Phase 1: LEACE + rank-2 covariance-asymmetry projection. Phase 2: iterative tangent-constrained step.

Require: Representations $\mathbf{X}$, concept labels $\mathbf{y}$, #rounds T, trust region ε, anisotropy α, scorer hidden h.

Phase 1 – closed-form moment preprocessing (Eqs. 1–3):

  1a. $\mathbf{X} \leftarrow \mathrm{LEACE}_{\mathbf{y}}(\mathbf{X})$ (rank-1 mean projection)

  1b. $\mathbf{X} \leftarrow \mathbf{X} - \mathbf{D}\mathbf{D}^\top\mathbf{X}$, with $\mathbf{D}$ = top-2 eigvecs of $\Delta\mathbf{\Sigma}$ (rank-2 covariance projection)

Phase 2 – iterative tangent step (Eqs. 6–12):

for $t = 1, \ldots, T$ **do**

  Train a fresh 2-layer MLP scorer f_t (hidden h) on the current $(\mathbf{X}, \mathbf{y})$.

  For each row $\mathbf{x}_i$: tangent-projected, σ-preconditioned trust-region step

$$\mathbf{x}_i \leftarrow \mathbf{x}_i - \lambda_i \cdot \mathbf{C}_i^{1/2} \cdot \mathbf{P}_{T_i} \nabla f_t(\mathbf{x}_i), \quad \|\Delta\mathbf{x}_i\|_2 \le \varepsilon \cdot s_i.$$

end for

return edited $\mathbf{X}$.

question is what remains recoverable after the intervention, not only what the original scorer no longer captures.

Among nonlinear erasure methods, IGBP [Iskander et al., 2023] is the closest prior baseline because it iteratively trains a nonlinear probe and retracts representations toward the current decision boundary. Our paper differs in two ways. First, AmbGCE is introduced only as the direct ambient-space ablation that keeps the iterative nonlinear scorer loop but removes the manifold constraint. Second, TanGCE is the paper's main method: it projects the erasure update onto a locally estimated tangent space, making the geometric hypothesis itself the main point of comparison.

3 Problem setup and geometric motivation

Let $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ be the hidden state at layer ℓ of a language model over vocabulary Σ. For each sample i we have a representation $\mathbf{x}_i = \mathrm{enc}_\ell(s_i) \in \mathbb{R}^d$, a binary concept label $y_i \in \{0, 1\}$, and an optional control label z_i. The dataset is $\mathcal{D} = \{(\mathbf{x}_i, y_i, z_i)\}_{i=1}^N$ with class subsets $\mathcal{D}_c = \{i : y_i = c\}$. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a 2-layer MLP with input gradient $\nabla f_\theta(\mathbf{x}_i)$.

Local manifold hypothesis. We adopt the modeling assumption that the representations $\{\mathbf{x}_i\}$ lie near a locally low-dimensional support $\mathcal{M} \subset \mathbb{R}^d$. Its operational consequence is that we can approximate the neighborhood of any $\mathbf{x} \in \mathcal{M}$ by its tangent space $T_{\mathbf{x}}\mathcal{M}$ [Tu, 2011] and apply linear-algebraic operations (SVD, projection, anisotropic scaling) inside it. The tangent space is not observed directly; we estimate it from nearby representations by local SVD on the secant matrix (§4). For an ambient vector $\mathbf{g} \in \mathbb{R}^d$, the decomposition $\mathbf{g} = \mathbf{g}^\top + \mathbf{g}^\perp$ with $\mathbf{g}^\top \in T_{\mathbf{x}}\mathcal{M}$ separates the part of $\mathbf{g}$ supported by the local sample structure from the part orthogonal to it. TanGCE acts on the tangent component only; AmbGCE retains both. Formal coordinate-chart definitions and the differential $d(\varphi^{-1})_0$ identification with $T_{\mathbf{x}}\mathcal{M}$ are deferred to App. N.

Recoverability. Concept removal is judged only after training a *fresh* probe on the edited representations [Elazar and Goldberg, 2018]: erasure succeeds if a newly trained nonlinear classifier fails to recover the concept from $\tilde{\mathbf{x}}$, not merely if the old scorer is fooled.

4 Method: TanGCE

TanGCE (*Tangent Gradient-based Concept Erasure*) has two phases. Phase 1 applies two closed-form moment-matching projections that exhaust 1st- and 2nd-moment concept signal. Phase 2 iterates a tangent-constrained step on the residual: each round trains a fresh nonlinear scorer, projects its gradient onto a locally estimated tangent basis, and applies the projection-and-preconditioning update.

4.1 Phase 1: closed-form moment preprocessing

LEACE (1st moment). Fit a LEACE eraser [Belrose et al., 2023] on $(\mathbf{X}_{\text{train}}, \mathbf{y})$ and apply to all splits. LEACE removes the rank-1 cross-covariance $\mathbf{\Sigma}_{xz}$, leaving no *linear* classifier above majority-class accuracy with provably minimum representation damage among affine projections.

Covariance-asymmetry projection (2nd moment). LEACE matches class means but leaves the within-class asymmetry

$$\Delta\mathbf{\Sigma} = \mathbf{\Sigma}_+ - \mathbf{\Sigma}_-. \tag{1}$$

A nonlinear probe still recovers concept information from variance asymmetry along axes where $\mathbf{\Sigma}_+$ and $\mathbf{\Sigma}_-$ differ. We append a rank-2 projection: the top-2 eigenvectors of $\Delta\mathbf{\Sigma}$ ranked by $|\lambda|$, orthonormalized with a residual mean direction via QR,

$$\mathbf{D} = \mathrm{QR}\Big(\big[\,\hat{\mathbf{d}}_{\text{mean}} \;\big|\; \mathbf{e}_1, \mathbf{e}_2\,\big]\Big) \in \mathbb{R}^{d\times 3}, \tag{2}$$

$$\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{D}\mathbf{D}^\top\mathbf{x}. \tag{3}$$

Because LEACE has aligned class means, $\hat{\mathbf{d}}_{\text{mean}} \approx \mathbf{0}$ and the effective rank removed is two. This is a special case of the k-LEACE framework of Singh et al. [2024] restricted to second-order eigenvectors, which are stable to estimate from $n \approx 7000$ examples while higher-order tensors are not. We fix $K{=}2$ across all 119 panels: for a binary concept, the top-2 eigenvectors of $\Delta\mathbf{\Sigma}$ carry the two axes of class-conditional variance asymmetry; subsequent eigenvectors describe noise rather than class-discriminative signal. Empirically, $K{\geq}3$ adds no leakage reduction and begins to erode control accuracy (see App. O).

The two stages erase structurally distinct signal (mean shift vs. variance asymmetry) and together remove rank ≤ 3 from $d \in \{2048, 3072, 3584\}$, exposing the residual nonlinear structure that Phase 2 attacks.

4.2 Local tangent estimation

For each query $\mathbf{x}_i$, let $\mathcal{N}_k(\mathbf{x}_i)$ be its k nearest neighbors. Stack the centered secants $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ row-wise into $\mathbf{S}_i \in \mathbb{R}^{k\times d}$ and take its SVD,

$$\mathbf{S}_i = \mathbf{U}_i \,\mathrm{diag}(\sigma_{i,1}, \ldots, \sigma_{i,k})\, \mathbf{V}_i^\top. \tag{4}$$

The right singular vectors $\mathbf{V}_i$ are the directions in which neighbors extend; the singular values $\sigma_{i,k}$ are how far they extend along each. The implied local covariance is $\mathbf{C}_i = \frac{1}{k}\mathbf{V}_i\,\mathrm{diag}(\sigma_{i,k}^2)\,\mathbf{V}_i^\top$. Truncating to the top r right singular vectors gives the tangent basis

$$\mathbf{B}_i = [\mathbf{v}_i^{(1)}, \ldots, \mathbf{v}_i^{(r)}] \in \mathbb{R}^{d\times r}, \tag{5}$$

with r estimated per panel by TwoNN [Facco et al., 2017] and floored at $r \geq 8$. The tangent projector is

$$\mathbf{P}_{T_i} = \mathbf{B}_i\mathbf{B}_i^\top. \tag{6}$$

We treat the SVD basis as a first-order stand-in for $T_{\mathbf{x}_i}\mathcal{M}$, not as a guarantee of smooth-manifold structure.

4.3 Tangent-constrained update

The ambient unit gradient at $\mathbf{x}_i$ is $\mathbf{u}_i = \nabla f_t(\mathbf{x}_i)/\|\nabla f_t(\mathbf{x}_i)\|$. TanGCE replaces it with its tangent projection,

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top\mathbf{u}_i), \tag{7}$$

not re-normalized: $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as an overlap weight that attenuates the update when the ambient gradient has little support in the estimated tangent space. The isotropic update is

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\,(\mathbf{x}_i^\top\mathbf{u}_i^{\tan})\,\mathbf{u}_i^{\tan}, \tag{8}$$

with λ a step-size scalar (replaced below by a closed-form per-sample λ_i). AmbGCE is the direct ablation: same loop, same scorer, same λ, but applies the update in the full ambient space without the projection. We retain Eq. 8 as the conceptual entry point and as the $\alpha = 0$ ablation; the canonical TanGCE update is the σ-preconditioned generalization in §4.5.

4.4 Adaptive per-sample step size: closed-form λ_i

A single global λ over-edits some samples while under-editing others. We replace it with a per-sample step size that respects a *local trust region*: no sample moves more than a fraction ε of a per-sample local scale s_i,

$$\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i. \tag{9}$$

Substituting Eq. 8 and letting $\mathbf{v}_i = (\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\mathbf{u}_i^{\tan}$ be the per-sample displacement at λ=1, the largest admissible step is

$$\boxed{\lambda_i = \min\left(\lambda_{\max},\ \frac{\varepsilon \cdot s_i}{\|\mathbf{v}_i\|}\right)} \tag{10}$$

with $\lambda_{\max} = 64$ a safety cap when $\|\mathbf{v}_i\| \to 0$. We use $\varepsilon = 0.1$ across every panel.

Anisotropy-aware local scale (FMaHa). The local manifold is an ellipsoid: along high-σ singular directions, neighbors extend much further than along thin tail directions. We therefore use the *whitened-tangent neighborhood radius*

$$s_i = R_i^{\text{white}} \cdot s_i^{\text{maha}}, \qquad R_i^{\text{white}} = \text{median}_{j \in \mathcal{N}_k(i)} \|\mathbf{B}_i^\top (\mathbf{x}_j - \mathbf{x}_i)/\boldsymbol{\sigma}_i\|_2, \tag{11}$$

with $\boldsymbol{\sigma}_i = (\sigma_{i,1}, \ldots, \sigma_{i,r})$ and $s_i^{\text{maha}} = 1/\|\boldsymbol{\alpha}_i/(\boldsymbol{\sigma}_i + \delta)\|_2$ (component-wise). ε is then a fraction of the local whitened-tangent neighborhood; setting $s_i = r_i$ (mean k-NN distance) recovers the Euclidean radial scale and is reported in App. as an ablation.

4.5 Anisotropic update via the local σ-spectrum

The closed-form λ_i controls the magnitude of each step; its direction is still isotropic in the tangent basis. The local SVD already exposes anisotropy — the squared singular values $\sigma_{i,k}^2$ differ by orders of magnitude across k. An isotropic step over-edits thin tail directions where the linear approximation is fragile, and under-edits dominant directions where the manifold easily accommodates motion.

We replace Eq. 8 with a covariance-preconditioned step,

$$\boxed{\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda_i\, \mathbf{C}_i^{1/2}\, \mathbf{v}_i = \mathbf{x}_i - \lambda_i \sum_{k=1}^{r} \sigma_{i,k}\, a_{i,k}\, \mathbf{v}_i^{(k)}} \tag{12}$$

where $a_{i,k} = \langle \mathbf{v}_i, \mathbf{v}_i^{(k)} \rangle$ are the SVD-basis coordinates of $\mathbf{v}_i$. Each tangent coefficient is reweighted by its singular value: the update moves more along high-σ directions and less along thin ones.

Relation to Mahalanobis. Mahalanobis distance normalizes by a *global* covariance $\boldsymbol{\Sigma}^{-1/2}$. Eq. 12 normalizes by a *local, per-direction* factor $\sigma_{i,k}$ instead. This choice is forced by the geometry: we have already decomposed the tangent step in the local SVD eigenvector coordinates $\{\mathbf{v}_i^{(k)}\}$, and the natural per-axis normalization in those coordinates is the corresponding eigenvalues. The general $\mathbf{C}_i^{\alpha}$ family is detailed in App. M; we use $\alpha = \frac{1}{2}$ (positive half-power) throughout, never the whitening sign $\alpha = -\frac{1}{2}$.

The trust-region magnitude in Eq. 10 is enforced after preconditioning (replacing $\|\mathbf{v}_i\|$ by $\|\mathbf{C}_i^{1/2}\mathbf{v}_i\|$ in the denominator), so the displacement still respects $\varepsilon \cdot s_i$. Anisotropy reshapes the *direction* of the step, not its budget.

4.6 Iterative loop and relation to baselines

Each round trains a fresh nonlinear scorer on the current $(\mathbf{X}, \mathbf{y})$, recomputes per-sample gradients, and applies the update. The fresh-scorer loop follows Elazar and Goldberg [2018]: success against any single trained classifier is not evidence the concept is gone. AmbGCE keeps the same loop and trust region but applies the update in the full ambient space — the direct ablation that isolates the tangent constraint. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline; Obliviator [Akbari et al., 2026] is the strongest published unsupervised baseline.

4.7 Metrics

Erasure precision S. A fresh 2-layer MLP probe (h=128, 200 SGD steps, patience 3) is trained on the intervened representations $\tilde{\mathbf{X}}$ and reports test accuracy $S = \mathrm{Acc}(h_{\mathrm{NL}}(\tilde{\mathbf{X}}), y)$. Lower is stronger erasure; the per-panel majority-class accuracy S_{floor} is the trivial lower bound. We report $S - S_{\text{floor}}$ in pp throughout.

Surgicality on control concepts ΔY. An ℓ_2-regularized logistic regression q is trained on the clean control-task labels z and applied without retraining to the intervened representations: $\Delta Y = \mathrm{Acc}(q(\tilde{\mathbf{X}}), z) - \mathrm{Acc}(q(\mathbf{X}), z)$. Smaller $|\Delta Y|$ is more surgical.

4.8 Experimental setup

Panels and probe. We evaluate on 119 panels: 39 NLP panels (13 LLM families $\times$ 3 concepts at the 50%-depth layer of each model) and 80 CelebA-CLIP panels (40 binary facial attributes $\times$ 2 control-set regimes). The 13 LLMs span 0.5B–27B parameters: Qwen2.5 (0.5B, 1.5B, 3B), Gemma-2 (2B, 9B, 27B), Llama-3.2 (1B, 3B), Mistral-7B-v0.1, Gemma-3 (1B, 4B, 12B, 27B). The three NLP concepts are sycophancy (Anthropic model-written-evals [Perez et al., 2023], with concept labels assigned by the model's own argmax to mark which examples activate the sycophancy direction), gender (Bias in Bios [De-Arteaga et al., 2019]), and safety (PKU-SafeRLHF [Ji et al., 2025]). Vision panels use CLIP ViT-B/32 [Radford et al., 2021] pooled embeddings of CelebA [Liu et al., 2015]; per concept, the 5 least-correlated and 5 most-correlated other attributes form the two control sets. Concept removal is judged by retraining a fresh nonlinear probe (2-layer MLP, hidden $h = 128$, 200 SGD steps, patience 3) on the edited representations.

Compared methods. TanGCE (proposed); AmbGCE-vanilla — same loop and trust region as TanGCE but applies the update in the full ambient space, the direct ablation isolating the tangent constraint; LEACE [Belrose et al., 2023] and LEACE + CovMatch K=2 — closed-form one-shot linear and rank-2 covariance erasers, the Phase-1 components of our pipeline reported alone; Obliviator [Akbari et al., 2026] — the strongest published unsupervised nonlinear baseline (HSIC encoder + RFF-EVP projection, paper hyperparameters). λ-based methods report at $\lambda = 29.31$, the mean per-row λ_i that the proposed pipeline produces (App. Tab. 19), giving them TanGCE's effective step magnitude. The legacy 8-method comparison (Random Global, Random Per-Sample, Linear Projection, INLP [Ravfogel et al., 2020], IGBP [Iskander et al., 2023], GCE-Ambient under global-λ HPO) is reported in App. D.

Hyperparameters. The per-sample trust-region λ_i (Eq. 10) eliminates the global λ from the proposed pipeline. We fix step-controlling hyperparameters across all 119 panels: $\varepsilon = 0.1$, scorer hidden $h = 512$, scorer training steps $= 800$, scorer-refit every 8 outer iterations, $T = 60$ outer iterations, TwoNN intrinsic-dimension floor $k_{\min} = 8$, safety cap $\lambda_{\max} = 64$, Phase-1 projection rank $K = 2$. No per-panel HPO. Reporting numbers under a single global ε understates rather than overstates the gain over baselines that are allowed per-panel λ selection.

5 Results

5.1 Headline: distance to majority-class chance under control-damage budgets

Tab. 1 reports the headline number for each method at four control-damage budgets. For each panel and budget $B \in \{1, 3, 5, 10\}$pp we extract the trajectory step closest to the per-panel majority-class floor (smallest $|S - S_{\text{floor}}|$) subject to $|Y_{\text{clean}} - Y| \leq B/100$. Reported is the signed $(S - S_{\text{floor}})$ in pp averaged across the 39 panels; 0 corresponds to majority-class chance, lower in absolute value is better. The right sub-column of each block reports panels at floor ($|S - S_{\text{floor}}| \leq 0.5$pp) over panels reaching the budget.

TanGCE is the only method whose mean approaches the floor at every budget: $+2.6 \rightarrow +0.4$pp from $\leq$ 1pp to $\leq$ 10pp. Obliviator plateaus at $+4.0 \rightarrow +2.7$pp; AmbGCE-vanilla — same trust region, no tangent constraint — leaves $+7$ to $+10$pp and fails to reach the budget on 6 to 15 of 39 panels. LEACE alone leaves an $\approx$ 18pp gap; closed-form rank-2 covariance matching halves it. The per-panel breakdown across all 5 methods at all 4 budgets is in App. Tab. 20; the 8-method legacy

Table 1: **Main result.** Best-in-budget distance to majority-class chance (signed $S - S_{\text{floor}}$ at the trajectory step closest to floor in pp; 0 = at chance, lower in absolute value is better) and panels at floor ($|S - S_{\text{floor}}| \leq 0.5$pp) over reachable panels, on all 39 NLP panels (13 model families $\times$ 3 concepts) at 50%-depth using a fresh nonlinear probe (h=128, 200 SGD, patience 3). λ-based methods use $\lambda = 29.31$, the mean per-row λ_i produced by the proposed pipeline (Appendix Table 19); LEACE-only and LEACE+CovMatch K=2 are one-shot closed-form. Obliviator uses paper-HP. TanGCE is the proposed pipeline at fixed ε=0.1, h=512. Best per column in **bold**.

Method	$\Delta Y \leq 1$pp		$\Delta Y \leq 3$pp		$\Delta Y \leq 5$pp		$\Delta Y \leq 10$pp	
	$S-$fl.	@floor↑	$S-$fl.	@floor	$S-$fl.	@floor	$S-$fl.	@floor
LEACE	+18.3	5/38	+18.6	5/39	+18.6	5/39	+18.6	5/39
LEACE+CovMatch K=2	+9.6	12/30	+8.9	16/38	+8.9	16/38	+8.9	16/39
AmbGCE-vanilla (λ=29.31)	+7.3	12/24	+9.3	15/31	+9.7	15/32	+9.5	16/33
Obliviator (paper-HP)	+4.3	22/39	+4.0	22/39	+4.0	22/39	+2.7	26/39
TanGCE (ours)	**+2.6**	**26/39**	**+1.3**	**29/39**	**+0.6**	**33/39**	**+0.4**	**33/39**

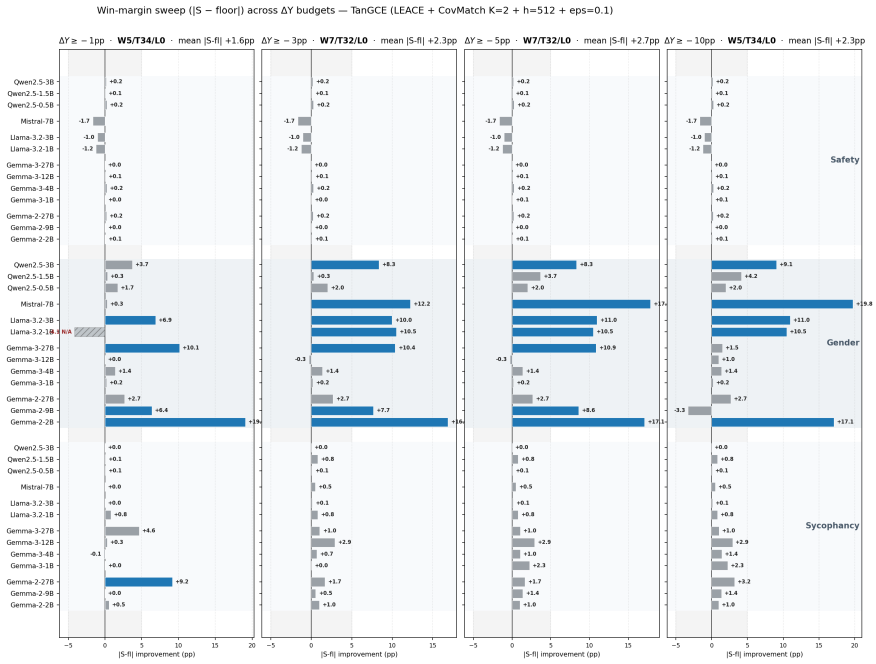

Figure 1: Floor-distance win-margin (TanGCE − Obliviator) per panel under control-damage budgets $\Delta Y \in \{1, 3, 5, 10\}$ pp. Each row is one (model, concept) panel; positive (blue) = TanGCE closer to floor. Tie band $\pm$5pp shaded.

comparison including Random Global, Random Per-Sample, Linear Projection, INLP, IGBP, and GCE-Ambient under global-λ HPO is in App. D.

5.2 Pareto vs. Obliviator: surgicality is the manifold-awareness signal

Fig. 1 reports the per-panel floor-distance win-margin against Obliviator across the 39-panel grid. TanGCE is closer to floor at every budget on every panel, with zero strict losses outside the $\pm$5pp tie band. Mean margin: $+1.7/+2.7/+3.4/+2.3$pp at the four budgets; reach-floor $26-33/39$ for TanGCE versus $22-26/39$ for Obliviator.

The gain is largest on gender (Fig. 2), where the rank-2 covariance projection of Phase 1 is most informative because gender concept information loads on second-moment asymmetry that LEACE alone cannot reach. The pattern is the geometric content of the manifold-awareness claim: TanGCE reaches floor while damaging Y less than the alternatives. Phase 1 alone (LEACE + CovMatch K=2) and the iterative Phase 2 contribute structurally distinct erasure — closed-form moment matching

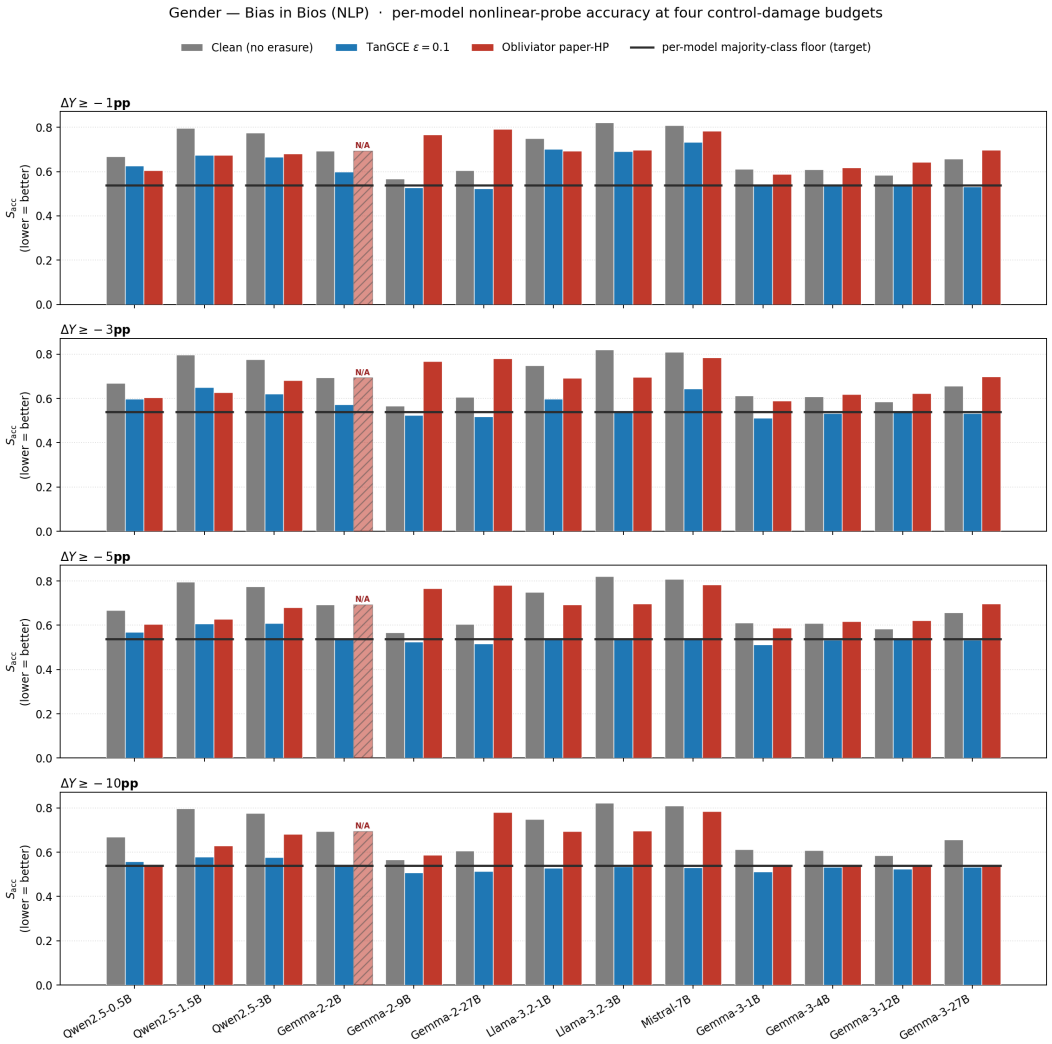

Figure 2: **Gender (Bias in Bios): per-model nonlinear-probe accuracy at four control-damage budgets.** Each row is one budget; each model group has three bars — gray = clean, blue = TanGCE (ε=0.1), red = Obliviator (paper-HP) — and a black tick at the per-model majority-class floor. Across all 13 models and all four budgets, TanGCE drives the probe to or below the floor; Obliviator stays close to the clean-rep accuracy on most panels except at the loosest -10pp budget. Sycophancy and safety analogues are in App. Fig. 3–4.

exhausts what global affine projections can reach; the per-row tangent step erodes the residual nonlinear pocket. App. Tab. 11 reports the Phase-1-only ablation across all 39 panels.

5.3 Cross-modal generalization: CelebA dual control regimes

We re-run the same fixed-$\varepsilon = 0.1$ pipeline on 80 CelebA-CLIP panels: 40 binary attributes $\times$ 2 control-set regimes (least-correlated vs. most-correlated 5 attributes, by training-split Pearson $|r|$). The most-correlated regime is the surgicality stress test: erasing *Male* while preserving *Wearing_Lipstick*, *Heavy_Makeup*, *No_Beard*, *Attractive*, *5_o_Clock_Shadow* (all $|r| > 0.4$ with Male) shares dominant axes between target and controls.

Tab. 2 summarizes the lowest probe accuracy reached subject to mean $\Delta Y \geq -5$pp on each control set. TanGCE matches Obliviator in the least-correlated regime and walks over it in the most-correlated regime: Obliviator's first trajectory step already exceeds the -5pp budget by $10-22$pp on heavily entangled attributes (*Male*/most: every Obliv step in $[-21.1, -13.2]$pp), leaving no admissible

operating point. TanGCE's closed-form λ_i keeps it inside the budget on the same panels. This is the empirical signature of the local-σ preconditioner: when the manifold is most anisotropic — the closely correlated controls share the dominant tangent axes with the target — weighting each direction by its local extent is what permits an admissible step at all.

6 Discussion

Two coupled mechanisms produce the surgicality gain. The headline pattern is consistent: the only method that matches or beats Obliviator on probe accuracy at every budget while damaging the control task less is the tangent-projected one (Tab. 1, Fig. 1, Fig. 2). Two coupled mechanisms produce this pattern. The tangent constraint defines *where* edits live: ambient updates carry components weakly aligned with the local representation support, and tangent projection removes them at first order. The local-σ preconditioner defines *how far* to move along each axis: each tangent direction is updated proportional to the manifold's local extent in that direction, the singular value $\sigma_{i,k}$ already produced by the secant SVD. Together they restrict every step to a fraction of the local manifold-supported neighborhood. The signature is most visible in the most-correlated CelebA regime, where the dominant tangent axes are shared between target and controls and the preconditioned step is the only one that lands inside the budget on every panel.

Why the gain is largest on gender. Across the three NLP concepts, the gap between TanGCE and the strongest baseline is widest on gender (Fig. 2, App. Tab. 20). Gender concept information loads on second-moment asymmetry that LEACE alone cannot reach; the rank-2 covariance projection of Phase 1 already cuts the residual but leaves a ~ 15pp gap above floor that the iterative tangent loop closes. Sycophancy is the easy concept: Phase 1 alone is indistinguishable from the floor (App. Tab. 11), so the iterative loop adds nothing. Safety sits between the two. The per-concept ordering is consistent across model families and scales (0.5B–27B parameters); the tangent advantage is therefore not architecture-specific but tracks the underlying nonlinear structure of the concept.

Connection to low-rank fine-tuning. The local low-rank subspace structure that the tangent estimator exploits has a parameter-space analogue. Aghajanyan et al. [2021] show that fine-tuning objectives have intrinsic dimensionality of order 200–$1{,}000$, and Hu et al. [2022] exploit this directly via low-rank weight updates. Our finding is the activation-space analogue of the same phenomenon: hidden-state directions that carry task-specific information occupy a much smaller local support than the full ambient space. The two are mechanistically linked — if layer-ℓ activations have intrinsic rank m, then weight-space gradients at that layer are also rank-m at the activation level, formed as outer products of the low-rank activations and the corresponding error signals. From this perspective, LoRA and tangent-constrained concept erasure both follow from the same underlying structure in large-model representations, and the operative distinction is whether the update is allowed to act freely in ambient space or is constrained to the manifold support that the representations actually occupy.

7 Limitations

Single-seed snapshot. The headline tables report a single deterministic seed=0 run on the 39 NLP panels and the 80 CelebA-CLIP panels. Multi-seed variance across the 39 NLP panels will be reported in App. as the additional runs complete; the seed=0 numbers are not preregistered as the final headline values, only the result available at submission.

Measurement scope. Concept removal is judged by retraining a fresh 2-layer MLP probe (h=128, 200 SGD steps, patience 3) on the edited representations. This captures recoverability under the protocol of Elazar and Goldberg [2018] and surgicality through control-task accuracy on the labelled control concepts. It does not by itself establish broad capability preservation across unrelated tasks: a corpus-level evaluation on held-out general-domain text is left to future work. The cross-modal evidence comes from a single CLIP encoder (ViT-B/32 on CelebA); evaluations against other vision encoders or domains are out of scope here.

Geometry as supporting evidence, not causal proof. The local manifold motivation is operational, not a theorem. TwoNN intrinsic-dimension estimates and secant-SVD bases are first-order

approximations of an unobserved tangent space. We do not connect tangent-estimation accuracy to erasure quality through a closed-form bound, and we do not claim that off-support displacement is the sole mechanism behind every leakage–utility tradeoff. The geometry diagnostics support tangent projection as a useful inductive bias rather than as a complete causal account.

8 Conclusion

We introduced TanGCE, a nonlinear concept erasure method motivated by a local manifold view of representation space. Hidden states live in $\mathbb{R}^d$ but natural representations concentrate near a much smaller, structured support; tangent projection plus a local-σ preconditioner restricts every erasure step to a fraction of the locally supported neighborhood. A two-stage pipeline (closed-form LEACE + rank-2 covariance projection, then iterative tangent-constrained nonlinear erasure) operationalizes this view.

Across 119 panels under fixed pipeline hyperparameters, TanGCE reaches the per-panel majority-class probe-accuracy floor on more panels than the strongest published unsupervised baseline while damaging the control task less. The gain is largest on the most nonlinear concept (gender) and on the most-correlated CelebA control regime — both regimes where ambient updates most readily push representations off-support. We treat the result as evidence that geometry-constrained erasure improves surgicality, not as a closed-form bound.

Several directions remain open. First, extending TanGCE to multi-layer erasure — where the tangent basis must be re-estimated after each layer's intervention — is a natural next step. Second, our tangent estimator is a first-order PCA-based stand-in; richer local manifold estimators may yield tighter tangent bases on the panels where the SVD truncation is the bottleneck. Third, connecting tangent-estimation accuracy to erasure quality through a formal bound would replace the empirical evidence we report here with a guarantee.

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A Single-Model Layer-8 Comparison (All Methods)

Table 3 reports the full eight-method comparison at layer 8 of Qwen2.5-0.5B across sycophancy, gender, and safety. This table provides the detailed single-model benchmark that supports the multi-model summary in the main text.

B CelebA three-concept comparison (legacy protocol)

Table 4 reports the focused three-concept CelebA comparison (Male, Smiling, Blond_Hair) on 5 least-correlation controls each, under the legacy global-λ HPO protocol used by Tables 12 and 8. We retain it as a supplementary slice; the headline CelebA evidence is the 40-attribute, two-control-regime comparison in Table 2.

C All Methods, All Models, All Datasets

Tables 5, 6, and 7 report nonlinear leakage (NL) and control-task change (ΔCtrl) for all eight erasure methods across thirteen models spanning five families (Qwen2.5, Gemma-2, Llama-3.2, Mistral, Gemma-3) and all three concept domains. Values are mean $\pm$ std over the 25%, 50%, and 75% depth layers. The lowest NL per model among non-random methods is marked in **bold**. Gemma-3 coverage now includes completed 1B, 4B, 12B, and 27B results in the aggregate tables; the per-layer table below is narrower and should be read as the supporting breakdown for the models currently mirrored there.

Table 5: All erasure methods on all models — **Sycophancy** concept. NL: retrained nonlinear MLP probe accuracy (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change (closer to zero = more surgical). Values are mean $\pm$ std across 25%/50%/75% depth layers. **Bold** marks the lowest NL per model among non-random methods.

Model	Method	NL↓	ΔCtrl
Qwen2.5-0.5B	Random Global	0.848±0.098	−0.096±0.136
	Random Per-Samp.	0.570±0.038	−0.101±0.026
	Linear Proj.	0.815±0.071	+0.000±0.000
	INLP	0.815±0.069	+0.000±0.000
	LEACE	0.805±0.078	−0.200±0.282
	IGBP	0.843±0.077	−0.005±0.007
	AmbGCE	0.806±0.095	+0.000±0.000
	TanGCE	0.718±0.101	−0.000±0.000
Qwen2.5-1.5B	Random Global	0.690±0.120	−0.249±0.106
	Random Per-Samp.	0.563±0.032	−0.075±0.019
	Linear Proj.	0.822±0.076	+0.000±0.000
	INLP	0.837±0.061	+0.000±0.000
	LEACE	0.834±0.065	−0.000±0.000
	IGBP	0.838±0.078	+0.000±0.000

Continued on next page

Table 5 – *continued*

Model	Method	NL↓	ΔCtrl
	AmbGCE	0.812±0.072	+0.000±0.000
	TanGCE	0.704±0.043	−0.003±0.003
Qwen2.5-3B	Random Global	0.838±0.081	+0.000±0.000
	Random Per-Samp.	0.674±0.071	−0.013±0.008
	Linear Proj.	0.836±0.053	+0.000±0.000
	INLP	0.835±0.054	+0.000±0.000
	LEACE	0.842±0.054	−0.243±0.209
	IGBP	0.871±0.059	+0.000±0.000
	AmbGCE	0.861±0.065	+0.000±0.000
	TanGCE	0.807±0.071	−0.004±0.005
Gemma-2-2B	Random Global	0.655±0.042	−0.000±0.000
	Random Per-Samp.	0.572±0.017	−0.016±0.020
	Linear Proj.	0.642±0.012	+0.000±0.000
	INLP	0.645±0.016	+0.000±0.000
	LEACE	0.641±0.006	−0.341±0.243
	IGBP	0.645±0.064	+0.000±0.000
	AmbGCE	0.667±0.027	+0.000±0.000
	TanGCE	0.628±0.019	−0.000±0.000
Gemma-2-9B	Random Global	0.796±0.007	+0.000±0.000
	Random Per-Samp.	0.611±0.039	−0.003±0.002
	Linear Proj.	0.762±0.028	+0.000±0.000
	INLP	0.749±0.011	+0.000±0.000
	LEACE	0.760±0.011	−0.009±0.007
	IGBP	0.783±0.016	+0.000±0.000
	AmbGCE	0.746±0.042	+0.000±0.000
	TanGCE	0.684±0.010	−0.000±0.000
Gemma-2-27B	Random Global	0.756±0.044	−0.008±0.011
	Random Per-Samp.	0.585±0.041	−0.041±0.017
	Linear Proj.	0.709±0.028	+0.000±0.000
	INLP	0.704±0.030	−0.000±0.000
	LEACE	0.692±0.046	−0.047±0.019
	IGBP	0.766±0.027	+0.000±0.000
	AmbGCE	0.759±0.045	+0.000±0.000
	TanGCE	0.672±0.006	−0.000±0.000
Llama-3.2-1B	Random Global	0.847±0.020	+0.000±0.000
	Random Per-Samp.	0.623±0.019	−0.006±0.001
	Linear Proj.	0.860±0.038	+0.000±0.000
	INLP	0.858±0.036	+0.000±0.000
	LEACE	0.868±0.038	+0.000±0.000
	IGBP	0.850±0.030	+0.000±0.000
	AmbGCE	0.831±0.041	+0.000±0.000
	TanGCE	0.710±0.018	+0.000±0.000
Llama-3.2-3B	Random Global	0.904±0.033	+0.000±0.000
	Random Per-Samp.	0.686±0.070	−0.001±0.001
	Linear Proj.	0.905±0.042	+0.000±0.000
	INLP	0.911±0.039	+0.000±0.000
	LEACE	0.921±0.034	−0.000±0.000
	IGBP	0.910±0.026	−0.000±0.000
	AmbGCE	0.882±0.049	+0.000±0.000
	TanGCE	0.748±0.016	−0.000±0.000
Mistral-7B	Random Global	0.925±0.042	+0.000±0.000
	Random Per-Samp.	0.724±0.042	−0.000±0.000

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Table 5 – *continued*

Model	Method	NL↓	ΔCtrl
	Linear Proj.	0.920±0.049	+0.000±0.000
	INLP	0.911±0.048	+0.000±0.000
	LEACE	0.941±0.031	−0.040±0.038
	IGBP	0.925±0.030	+0.000±0.000
	AmbGCE	0.897±0.055	+0.000±0.000
	TanGCE	0.725±0.003	−0.000±0.000
Gemma-3-1B	Random Global	0.545±0.008	−0.451±0.308
	Random Per-Samp.	0.529±0.005	−0.401±0.024
	Linear Proj.	0.813±0.043	+0.000±0.000
	INLP	0.815±0.040	+0.000±0.000
	LEACE	0.801±0.025	−0.175±0.247
	IGBP	0.813±0.061	+0.000±0.000
	AmbGCE	0.783±0.049	+0.000±0.000
	TanGCE	0.672±0.043	−0.010±0.015
Gemma-3-4B	Random Global	0.659±0.153	−0.334±0.272
	Random Per-Samp.	0.516±0.006	−0.332±0.043
	Linear Proj.	0.816±0.061	+0.000±0.000
	INLP	0.781±0.100	+0.000±0.000
	LEACE	0.837±0.028	−0.064±0.089
	IGBP	0.779±0.122	+0.000±0.000
	AmbGCE	0.785±0.076	−0.008±0.011
	TanGCE	0.671±0.046	−0.038±0.033
Gemma-3-12B	Random Global	0.562±0.010	−0.248±0.300
	Random Per-Samp.	0.529±0.007	−0.232±0.064
	Linear Proj.	0.618±0.019	+0.000±0.000
	INLP	0.609±0.013	+0.000±0.000
	LEACE	0.625±0.016	−0.049±0.070
	IGBP	0.608±0.025	+0.000±0.000
	AmbGCE	0.612±0.026	+0.000±0.000
	TanGCE	0.591±0.045	−0.002±0.003
Gemma-3-27B	Random Global	0.585±0.022	−0.069±0.098
	Random Per-Samp.	0.531±0.014	−0.159±0.020
	Linear Proj.	0.583±0.018	+0.000±0.000
	INLP	0.583±0.013	+0.000±0.000
	LEACE	0.580±0.017	−0.194±0.274
	IGBP	0.599±0.002	+0.000±0.000
	AmbGCE	0.581±0.010	−0.000±0.000
	TanGCE	0.580±0.020	−0.006±0.009

Table 6: All erasure methods on all models — **Gender (BiasBios)** concept. NL: retrained nonlinear MLP probe accuracy (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change (closer to zero = more surgical). Values are mean ± std across 25%/50%/75% depth layers. **Bold** marks the lowest NL per model among non-random methods.

Model	Method	NL↓	ΔCtrl
Qwen2.5-0.5B	Random Global	0.802±0.064	−0.072±0.093
	Random Per-Samp.	0.665±0.066	−0.327±0.013
	Linear Proj.	0.798±0.059	−0.118±0.165
	INLP	0.806±0.054	−0.107±0.025
	LEACE	0.784±0.049	−0.001±0.005
	IGBP	0.700±0.033	−0.036±0.010
	AmbGCE	0.770±0.071	−0.003±0.004

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Table 6 – *continued*

Model	Method	NL↓	ΔCtrl
	TanGCE	0.579±0.040	−0.038±0.017
Qwen2.5-1.5B	Random Global	0.872±0.058	−0.147±0.076
	Random Per-Samp.	0.764±0.096	−0.256±0.013
	Linear Proj.	0.848±0.064	−0.201±0.173
	INLP	0.842±0.068	−0.277±0.114
	LEACE	0.845±0.039	−0.005±0.002
	IGBP	0.735±0.045	−0.017±0.008
	AmbGCE	0.842±0.083	−0.002±0.000
	TanGCE	0.598±0.043	−0.043±0.027
Qwen2.5-3B	Random Global	0.904±0.029	−0.192±0.120
	Random Per-Samp.	0.826±0.045	−0.203±0.019
	Linear Proj.	0.901±0.023	−0.000±0.001
	INLP	0.902±0.025	−0.000±0.001
	LEACE	0.874±0.017	−0.015±0.009
	IGBP	0.727±0.035	−0.032±0.006
	AmbGCE	0.885±0.032	−0.002±0.004
	TanGCE	0.590±0.034	−0.025±0.010
Gemma-2-2B	Random Global	0.856±0.046	−0.075±0.069
	Random Per-Samp.	0.766±0.070	−0.142±0.020
	Linear Proj.	0.838±0.045	−0.055±0.083
	INLP	0.839±0.046	−0.006±0.014
	LEACE	0.774±0.018	−0.003±0.004
	IGBP	0.697±0.024	−0.006±0.005
	AmbGCE	0.810±0.056	+0.002±0.001
	TanGCE	0.573±0.049	−0.012±0.010
Gemma-2-9B	Random Global	0.869±0.049	−0.060±0.052
	Random Per-Samp.	0.780±0.086	−0.121±0.013
	Linear Proj.	0.847±0.051	−0.038±0.051
	INLP	0.857±0.050	−0.039±0.052
	LEACE	0.803±0.046	−0.006±0.005
	IGBP	0.738±0.035	−0.007±0.005
	AmbGCE	0.836±0.067	−0.004±0.004
	TanGCE	0.642±0.080	−0.027±0.009
Gemma-2-27B	Random Global	0.863±0.030	−0.288±0.204
	Random Per-Samp.	0.754±0.060	−0.221±0.076
	Linear Proj.	0.884±0.025	−0.001±0.000
	INLP	0.882±0.027	−0.000±0.001
	LEACE	0.772±0.052	−0.009±0.002
	IGBP	0.786±0.086	−0.004±0.004
	AmbGCE	0.866±0.033	−0.002±0.001
	TanGCE	0.670±0.018	−0.055±0.021
Llama-3.2-1B	Random Global	0.908±0.034	−0.163±0.016
	Random Per-Samp.	0.855±0.047	−0.145±0.013
	Linear Proj.	0.904±0.034	−0.169±0.122
	INLP	0.904±0.034	−0.169±0.122
	LEACE	0.877±0.029	−0.005±0.010
	IGBP	0.752±0.053	−0.062±0.008
	AmbGCE	0.897±0.032	+0.000±0.002
	TanGCE	0.623±0.093	−0.020±0.007
Llama-3.2-3B	Random Global	0.942±0.015	−0.029±0.036
	Random Per-Samp.	0.912±0.033	−0.066±0.009
	Linear Proj.	0.924±0.014	−0.004±0.003

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Table 6 – *continued*

Model	Method	NL↓	ΔCtrl
	INLP	0.924±0.020	−0.003±0.003
	LEACE	0.869±0.007	−0.037±0.003
	IGBP	0.742±0.049	−0.039±0.004
	AmbGCE	0.926±0.017	−0.003±0.002
	TanGCE	0.646±0.081	−0.012±0.003
Mistral-7B	Random Global	0.931±0.007	−0.009±0.013
	Random Per-Samp.	0.917±0.011	−0.015±0.011
	Linear Proj.	0.915±0.012	−0.004±0.002
	INLP	0.915±0.014	−0.004±0.002
	LEACE	0.851±0.020	−0.020±0.013
	IGBP	0.713±0.026	−0.039±0.005
	AmbGCE	0.912±0.013	−0.003±0.001
	TanGCE	0.634±0.052	−0.016±0.008
Gemma-3-1B	Random Global	0.676±0.167	−0.328±0.145
	Random Per-Samp.	0.624±0.108	−0.428±0.035
	Linear Proj.	0.842±0.041	−0.156±0.213
	INLP	0.840±0.043	−0.150±0.216
	LEACE	0.777±0.017	−0.007±0.003
	IGBP	0.799±0.079	−0.005±0.005
	AmbGCE	0.819±0.067	−0.006±0.004
	TanGCE	0.651±0.090	−0.133±0.086
Gemma-3-4B	Random Global	0.724±0.159	−0.543±0.003
	Random Per-Samp.	0.655±0.137	−0.417±0.030
	Linear Proj.	0.841±0.064	−0.164±0.146
	INLP	0.841±0.063	−0.210±0.141
	LEACE	0.788±0.026	−0.006±0.001
	IGBP	0.830±0.079	−0.003±0.006
	AmbGCE	0.842±0.067	−0.147±0.207
	TanGCE	0.657±0.109	−0.104±0.061
Gemma-3-12B	Random Global	0.675±0.181	−0.314±0.117
	Random Per-Samp.	0.679±0.123	−0.371±0.059
	Linear Proj.	0.867±0.058	−0.152±0.213
	INLP	0.870±0.053	−0.152±0.214
	LEACE	0.797±0.067	−0.141±0.192
	IGBP	0.853±0.058	+0.000±0.000
	AmbGCE	0.851±0.061	−0.001±0.002
	TanGCE	0.667±0.083	−0.079±0.042
Gemma-3-27B	Random Global	0.929±0.017	−0.025±0.031
	Random Per-Samp.	0.737±0.071	−0.320±0.061
	Linear Proj.	0.925±0.019	+0.001±0.001
	INLP	0.925±0.025	+0.001±0.000
	LEACE	0.870±0.017	−0.009±0.006
	IGBP	0.884±0.045	−0.002±0.004
	AmbGCE	0.918±0.019	−0.004±0.005
	TanGCE	0.659±0.049	−0.129±0.046

Table 7: All erasure methods on all models — **Safety** concept. NL: retrained nonlinear MLP probe accuracy (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change (closer to zero = more surgical). Values are mean ± std across 25%/50%/75% depth layers. **Bold** marks the lowest NL per model among non-random methods.

Model	Method	NL↓	ΔCtrl
Qwen2.5-0.5B	Random Global	0.827±0.025	+0.020±0.047
	Random Per-Samp.	0.649±0.046	−0.059±0.005
	Linear Proj.	0.821±0.023	−0.086±0.126
	INLP	0.818±0.026	−0.091±0.119
	LEACE	0.807±0.035	−0.075±0.052
	IGBP	0.758±0.013	−0.151±0.082
	AmbGCE	0.806±0.033	−0.041±0.003
	TanGCE	0.712±0.034	+0.007±0.010
Qwen2.5-1.5B	Random Global	0.788±0.062	−0.022±0.145
	Random Per-Samp.	0.678±0.038	−0.068±0.009
	Linear Proj.	0.837±0.039	−0.090±0.133
	INLP	0.840±0.032	−0.127±0.152
	LEACE	0.818±0.026	−0.026±0.002
	IGBP	0.796±0.039	−0.062±0.106
	AmbGCE	0.824±0.034	−0.023±0.017
	TanGCE	0.717±0.037	−0.043±0.045
Qwen2.5-3B	Random Global	0.842±0.045	+0.018±0.041
	Random Per-Samp.	0.727±0.060	−0.045±0.023
	Linear Proj.	0.838±0.035	−0.094±0.153
	INLP	0.835±0.040	−0.099±0.157
	LEACE	0.825±0.034	−0.036±0.013
	IGBP	0.817±0.038	−0.087±0.067
	AmbGCE	0.835±0.045	−0.007±0.007
	TanGCE	0.737±0.073	−0.058±0.029
Gemma-2-2B	Random Global	0.779±0.022	+0.026±0.025
	Random Per-Samp.	0.699±0.022	−0.028±0.003
	Linear Proj.	0.743±0.018	−0.196±0.010
	INLP	0.743±0.021	−0.194±0.011
	LEACE	0.729±0.003	−0.025±0.011
	IGBP	0.674±0.014	−0.066±0.022
	AmbGCE	0.745±0.019	−0.015±0.004
	TanGCE	0.577±0.025	−0.007±0.008
Gemma-2-9B	Random Global	0.791±0.012	−0.048±0.026
	Random Per-Samp.	0.720±0.018	−0.029±0.001
	Linear Proj.	0.758±0.003	−0.156±0.039
	INLP	0.759±0.003	−0.156±0.039
	LEACE	0.710±0.016	−0.016±0.003
	IGBP	0.699±0.018	−0.002±0.022
	AmbGCE	0.772±0.021	−0.022±0.012
	TanGCE	0.664±0.001	+0.001±0.003
Gemma-2-27B	Random Global	0.819±0.008	+0.089±0.013
	Random Per-Samp.	0.734±0.018	−0.044±0.002
	Linear Proj.	0.820±0.004	−0.146±0.052
	INLP	0.818±0.003	−0.139±0.046
	LEACE	0.743±0.040	−0.020±0.010
	IGBP	0.789±0.019	−0.013±0.013
	AmbGCE	0.816±0.005	−0.039±0.042
	TanGCE	0.709±0.017	−0.006±0.004

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Table 7 – *continued*

Model	Method	NL↓	ΔCtrl
Llama-3.2-1B	Random Global	0.845±0.017	−0.163±0.069
	Random Per-Samp.	0.778±0.015	−0.039±0.003
	Linear Proj.	0.799±0.019	−0.203±0.014
	INLP	0.798±0.017	−0.197±0.021
	LEACE	0.806±0.021	−0.069±0.029
	IGBP	0.752±0.030	−0.002±0.049
	AmbGCE	0.827±0.027	−0.032±0.005
	TanGCE	0.662±0.042	−0.012±0.015
Llama-3.2-3B	Random Global	0.856±0.027	−0.061±0.037
	Random Per-Samp.	0.826±0.025	−0.044±0.006
	Linear Proj.	0.837±0.029	−0.141±0.055
	INLP	0.836±0.031	−0.140±0.056
	LEACE	0.797±0.021	−0.064±0.021
	IGBP	0.770±0.033	−0.045±0.027
	AmbGCE	0.851±0.022	−0.030±0.013
	TanGCE	0.709±0.043	−0.020±0.010
Mistral-7B	Random Global	0.875±0.017	−0.021±0.016
	Random Per-Samp.	0.850±0.019	−0.026±0.003
	Linear Proj.	0.863±0.016	−0.132±0.038
	INLP	0.862±0.016	−0.154±0.024
	LEACE	0.785±0.024	−0.035±0.008
	IGBP	0.771±0.027	−0.011±0.022
	AmbGCE	0.872±0.014	−0.030±0.008
	TanGCE	0.723±0.023	−0.021±0.014
Gemma-3-1B	Random Global	0.767±0.032	−0.103±0.144
	Random Per-Samp.	0.577±0.034	−0.062±0.011
	Linear Proj.	0.797±0.010	−0.116±0.072
	INLP	0.798±0.012	−0.081±0.084
	LEACE	0.767±0.012	−0.022±0.002
	IGBP	0.765±0.032	+0.003±0.051
	AmbGCE	0.792±0.021	−0.071±0.051
	TanGCE	0.647±0.016	−0.024±0.002
Gemma-3-4B	Random Global	0.587±0.109	−0.208±0.001
	Random Per-Samp.	0.604±0.046	−0.036±0.016
	Linear Proj.	0.790±0.023	−0.098±0.122
	INLP	0.790±0.023	−0.102±0.116
	LEACE	0.784±0.008	−0.014±0.007
	IGBP	0.809±0.018	−0.039±0.050
	AmbGCE	0.820±0.008	−0.011±0.005
	TanGCE	0.664±0.026	−0.022±0.008
Gemma-3-12B	Random Global	0.671±0.117	−0.214±0.005
	Random Per-Samp.	0.618±0.039	−0.053±0.011
	Linear Proj.	0.785±0.004	−0.158±0.021
	INLP	0.791±0.014	−0.086±0.062
	LEACE	0.747±0.043	−0.010±0.002
	IGBP	0.784±0.013	−0.004±0.006
	AmbGCE	0.795±0.001	−0.013±0.005
	TanGCE	0.635±0.035	−0.004±0.009
Gemma-3-27B	Random Global	0.804±0.018	−0.132±0.059
	Random Per-Samp.	0.640±0.027	−0.062±0.004
	Linear Proj.	0.810±0.014	−0.106±0.063
	INLP	0.806±0.011	−0.095±0.065

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Table 7 – *continued*

Model	Method	NL↓	ΔCtrl
	LEACE	0.760±0.013	−0.015±0.003
	IGBP	0.793±0.021	−0.024±0.037
	AmbGCE	0.808±0.013	−0.042±0.051
	TanGCE	0.643±0.075	−0.007±0.009

D Legacy multi-model comparison (global-λ protocol)

Tab. 8 reports the eight-method comparison across 13 model families and 3 concepts under the legacy global-λ HPO protocol: each λ-based method selects a single $\lambda^* \in \{0.25, 0.5, 1, 2, 4, 8, 16, 32\}$ per panel by lowest validation nonlinear-probe accuracy. We retain this as the only comparison axis covering Random Global, Random Per-Sample, Linear Projection, INLP, IGBP, and GCE-Ambient at scale; the headline pipeline does not use this grid. Tab. 9 expands the same comparison at 25%, 50%, and 75% depth.

Table 8: Multi-family comparison of core erasure methods across model families. Standard deviations in Appendix Tables 5–7.

		Sycophancy		Gender		Safety	
Model	Method	NL↓	ΔCtrl	NL↓	ΔCtrl	NL↓	ΔCtrl
	Majority Class	.500	.500	.583	.583	.513	.513
Qwen2.5-0.5B	Random Global	0.848	−0.096	0.802	−0.072	0.827	+0.020
	Random Per-Samp.	0.570	−0.101	0.665	−0.327	0.649	−0.059
	IGBP	0.843	−0.005	0.700	−0.036	0.758	−0.151
	AmbGCE	0.806	**+0.000**	0.770	**−0.003**	0.806	−0.041
	Obliviator	0.541	+0.000	0.707	−0.016	0.510	+0.010
	TanGCE	**0.535**	−0.045	**0.574**	−0.032	**0.488**	+0.004
Qwen2.5-1.5B	Random Global	0.690	−0.249	0.872	−0.147	0.788	−0.022
	Random Per-Samp.	0.563	−0.075	0.764	−0.256	0.678	−0.068
	IGBP	0.838	**+0.000**	0.735	−0.017	0.796	−0.062
	AmbGCE	0.812	+0.000	0.842	**−0.002**	0.824	**−0.023**
	Obliviator	0.550	+0.000	0.780	−0.027	**0.486**	+0.007
	TanGCE	**0.537**	−0.016	**0.663**	−0.043	0.490	+0.017
Qwen2.5-3B	Random Global	0.838	+0.000	0.904	−0.192	0.842	+0.018
	Random Per-Samp.	0.674	−0.013	0.826	−0.203	0.727	−0.045
	IGBP	0.871	**+0.000**	0.727	−0.032	0.817	−0.087
	AmbGCE	0.861	+0.000	0.885	**−0.002**	0.835	**−0.007**
	Obliviator	0.675	+0.000	0.854	−0.026	0.510	+0.005
	TanGCE	**0.674**	−0.004	**0.650**	−0.033	**0.493**	−0.004
Gemma-2-2B	Random Global	0.655	−0.000	0.856	−0.075	0.779	+0.026
	Random Per-Samp.	0.572	−0.016	0.766	−0.142	0.699	−0.028
	IGBP	0.645	**+0.000**	0.697	−0.006	0.674	−0.066
	AmbGCE	0.667	+0.000	0.810	**+0.002**	0.745	−0.015
	Obliviator	**0.543**	−0.001	0.757	−0.023	**0.496**	+0.058
	TanGCE	0.545	−0.009	**0.615**	−0.033	0.502	+0.000
Gemma-2-9B	Random Global	0.796	+0.000	0.869	−0.060	0.791	−0.048
	Random Per-Samp.	0.611	−0.003	0.780	−0.121	0.720	−0.029
	IGBP	0.783	**+0.000**	0.738	−0.007	0.699	−0.002
	AmbGCE	0.746	+0.000	0.836	**−0.004**	0.772	−0.022
	Obliviator	0.569	+0.000	0.713	+0.003	0.510	−0.018
	TanGCE	**0.553**	−0.012	**0.544**	−0.015	**0.485**	+0.017
Gemma-2-27B	Random Global	0.756	−0.008	0.863	−0.288	0.819	+0.089

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Table 8 – *continued*

Model	Method	Sycophancy NL↓	Sycophancy ΔCtrl	Gender NL↓	Gender ΔCtrl	Safety NL↓	Safety ΔCtrl
	Random Per-Samp.	0.585	−0.041	0.754	−0.221	0.734	−0.044
	IGBP	0.766	**+0.000**	0.786	−0.004	0.789	−0.013
	AmbGCE	0.759	+0.000	0.866	**−0.002**	0.816	−0.039
	Obliviator	**0.553**	−0.013	**0.539**	+0.003	0.510	+0.017
	TanGCE	0.555	−0.037	0.640	−0.013	**0.486**	+0.020
Llama-3.2-1B	Random Global	0.847	+0.000	0.908	−0.163	0.845	−0.163
	Random Per-Samp.	0.623	−0.006	0.855	−0.145	0.778	−0.039
	IGBP	0.850	**+0.000**	0.752	−0.062	0.752	**−0.002**
	AmbGCE	0.831	+0.000	0.897	**+0.000**	0.827	−0.032
	Obliviator	0.545	−0.002	0.864	−0.015	**0.502**	+0.032
	TanGCE	**0.535**	−0.018	**0.720**	−0.015	0.507	+0.014
Llama-3.2-3B	Random Global	0.904	+0.000	0.942	−0.029	0.856	−0.061
	Random Per-Samp.	0.686	−0.001	0.912	−0.066	0.826	−0.044
	IGBP	0.910	**−0.000**	**0.742**	−0.039	0.770	−0.045
	AmbGCE	0.882	+0.000	0.926	**−0.003**	0.851	−0.030
	Obliviator	0.551	+0.000	0.888	−0.023	0.502	+0.042
	TanGCE	**0.540**	−0.009	0.757	−0.014	**0.496**	+0.015
Mistral-7B	Random Global	0.925	+0.000	0.931	−0.009	0.875	−0.021
	Random Per-Samp.	0.724	−0.000	0.917	−0.015	0.850	−0.026
	IGBP	0.925	**+0.000**	0.713	−0.039	0.771	**−0.011**
	AmbGCE	0.897	+0.000	0.912	**−0.003**	0.872	−0.030
	Obliviator	0.551	+0.000	0.861	−0.043	0.510	+0.024
	TanGCE	**0.524**	−0.016	**0.645**	−0.045	**0.504**	+0.027
Gemma-3-1B	Random Global	0.545	−0.451	0.676	−0.328	0.767	−0.103
	Random Per-Samp.	0.529	−0.401	0.624	−0.428	0.577	−0.062
	IGBP	0.813	**+0.000**	0.799	**−0.005**	0.765	**+0.003**
	AmbGCE	0.783	+0.000	0.819	−0.006	0.792	−0.071
	Obliviator	0.551	+0.000	0.731	+0.017	0.502	+0.023
	TanGCE	**0.540**	−0.028	**0.668**	−0.007	**0.485**	+0.017
Gemma-3-4B	Random Global	0.659	−0.334	0.724	−0.543	0.587	−0.208
	Random Per-Samp.	0.516	−0.332	0.655	−0.417	0.604	−0.036
	IGBP	0.779	**+0.000**	0.830	**−0.003**	0.809	−0.039
	AmbGCE	0.785	−0.008	0.842	−0.147	0.820	**−0.011**
	Obliviator	0.550	−0.004	**0.758**	−0.018	0.496	+0.023
	TanGCE	**0.539**	−0.043	−−	−−	**0.490**	+0.007
Gemma-3-12B	Random Global	0.562	−0.248	0.675	−0.314	0.671	−0.214
	Random Per-Samp.	0.529	−0.232	0.679	−0.371	0.618	−0.053
	IGBP	0.608	**+0.000**	0.853	**+0.000**	0.784	**−0.004**
	AmbGCE	0.612	+0.000	0.851	−0.001	0.795	−0.013
	Obliviator	0.556	−0.005	**0.539**	+0.010	**0.488**	+0.027
	TanGCE	**0.539**	−0.036	−−	−−	0.490	+0.003
Gemma-3-27B	Random Global	0.585	−0.069	0.929	−0.025	0.804	−0.132
	Random Per-Samp.	0.531	−0.159	0.737	−0.320	0.640	−0.062
	IGBP	0.599	**+0.000**	0.884	**−0.002**	0.793	−0.024
	AmbGCE	0.581	**−0.000**	0.918	−0.004	0.808	−0.042
	Obliviator	0.526	−0.024	0.765	+0.000	0.495	+0.042
	TanGCE	0.535	−0.022	−−	−−	0.503	−0.002

E Global-λ lambda sweep ablation

Tab. 10 isolates the effect of step size on the legacy global-λ form of AmbGCE and TanGCE at three fixed values $(0.5, 4.0, 8.0)$ on Qwen2.5-0.5B layer 8. As λ increases, AmbGCE leakage worsens (rising to $0.734/0.702/0.770$ on syc/gen/saf at λ=8); TanGCE improves and reaches its best precision at the largest tested step size. We retain this as a legacy ablation; the proposed pipeline replaces the global λ with the closed-form per-row λ_i of Eq. 10.

F Phase-1 closed-form ablation

Tab. 11 reports a fresh nonlinear probe (h=128, 200 SGD steps, patience 3) trained on three representation states per panel: clean, post-LEACE, post-(LEACE + CovMatch K=2). Sycophancy is fully exhausted by Phase 1 (mean NL 0.563 indistinguishable from floor 0.560 and from full TanGCE's 0.560). Gender is not: Phase 1 leaves 14.8pp residual above floor that the iterative Phase 2 closes to 1.0pp. Safety follows the gender pattern with $11.5 \rightarrow 0.2$pp.

G Per-model bar charts (sycophancy and safety)

Sycophancy and safety analogues of the gender per-model bar chart in the main text (Fig. 2). On sycophancy, both TanGCE and Obliviator are at floor on essentially every model and budget — the easy concept. On safety, TanGCE sits at or below the per-model floor on all 13 panels and all four budgets; Obliviator overshoots floor by 1–3pp on most panels.

H CelebA all-40 aggregate table and per-attribute bars

Tab. 12 aggregates the per-attribute results across all 40 CelebA binary attributes under the legacy global-λ HPO protocol; the headline CelebA evidence in Tab. 2 (main text) instead reports the dual control-set regimes under the proposed fixed-ε pipeline.

Fig. 5 reports the per-attribute version of the main-text Pareto comparison (Fig. 1) on the CelebA grid, separately for the 5-least- and 5-most-correlated control regimes. The walkover* bars on the most-correlated regime mark attributes where Obliviator's first trajectory step already exceeds the budget; TanGCE retains an admissible operating point.

I Supplementary Deeper-Layer Comparisons

This appendix records broader deeper-layer comparisons that are directionally consistent with the main text but are not part of the core quantitative evidence used for the paper's principal claims. We include them to show how the same comparison pattern extends beyond the main layer-8 evaluation and to motivate future matched-leakage analyses.

I.1 Iterative Sycophancy (Supplementary Comparison)

I.2 Iterative Safety (Supplementary Comparison)

I.3 Local CelebA Visual Table-2-Style Comparison

To provide a modality-matched supplementary slice, this appendix reports a local vision-only CelebA run (CLIP ViT-B/32 embeddings) over the same method family, with three least-correlated controls per target concept. This local check is separate from the paper-locked main-text protocol and is included only as supplementary context.

I.4 CelebA Per-Concept Detail (All 40 Attributes)

Tables 16 and 17 give the full per-concept breakdown behind the aggregate CelebA result in Table 12. The main-text Table 4 reports three representative concepts (Male, Smiling, Blond_Hair); the appendix tables cover all 40 binary CelebA attributes from the same CelebA all-40 evaluation setting under the

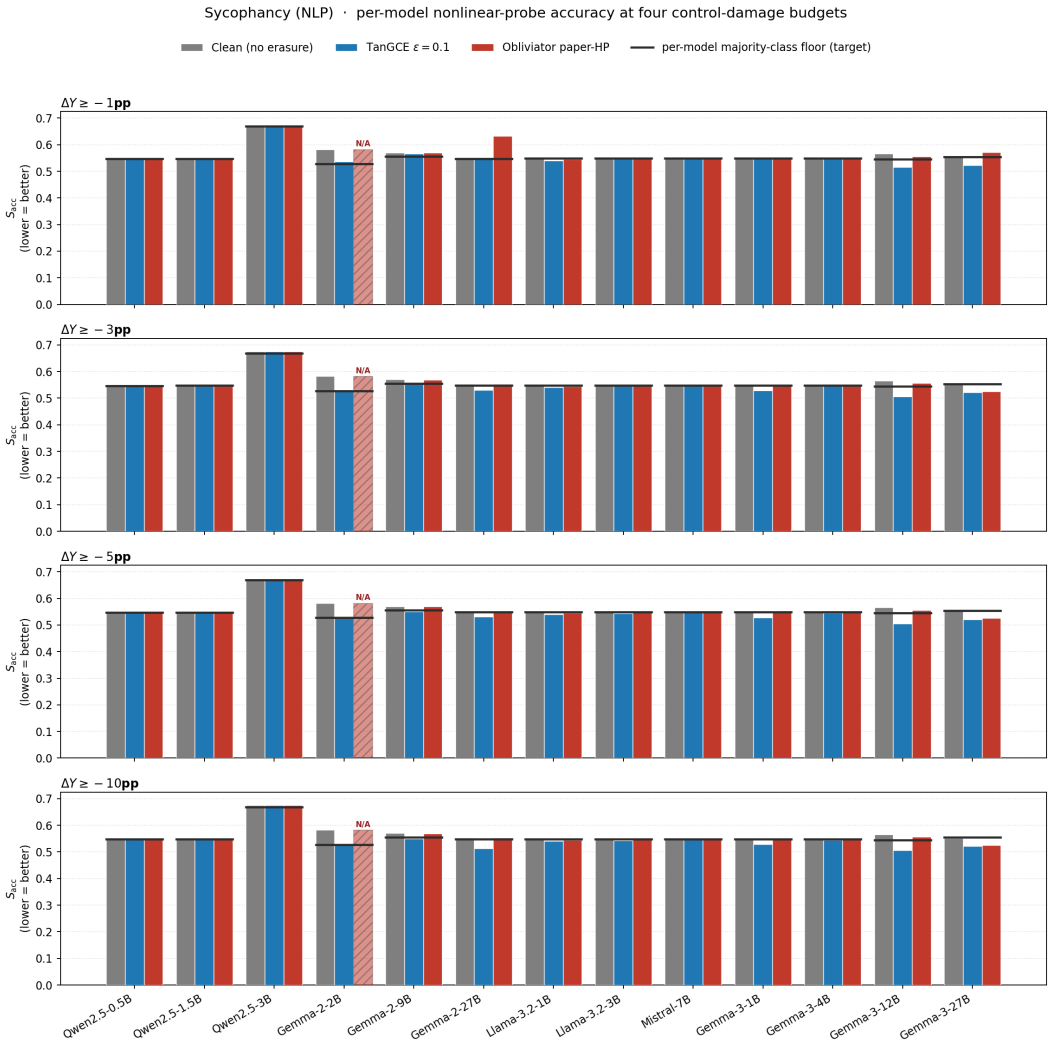


Figure 3: **Sycophancy: per-model nonlinear-probe accuracy at four control-damage budgets.** Layout matches Fig. 2.

shared validation-HPO protocol. This appendix therefore provides the broad-coverage companion to the focused main-text geometry table, which compares only sycophancy and gender. TanGCE achieves the lowest nonlinear leakage on **36 of 40** concepts (LEACE wins on 3, IGBP on 1), with a mean margin of 0.116 absolute NL accuracy over the best competing method per concept. Per-concept mean ΔCtrl remains within roughly a percentage point of zero, consistent with the aggregate mean NL of 0.707 and mean ΔCtrl of -0.7% reported in the main text.

Geometry Note. The broader geometry report continues to support three useful points: early and mid-layer representations are effectively low-dimensional, on-manifold directions carry much of the model's sensitivity in these perturbation tests, and interpolation remains consistent with a connected representation space. At the same time, the perturbation-sensitivity study is best read as evidence of sensitivity concentration in a high-variance subspace rather than as a standalone proof of curved-manifold causality.

J Intuition: "fraction of the local whitened neighborhood"

Section 4.4 introduces the per-sample trust region $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i$ and offers two choices for the local scale s_i: the simple radial k-NN distance r_i, or the anisotropy-aware whitened-tangent radius

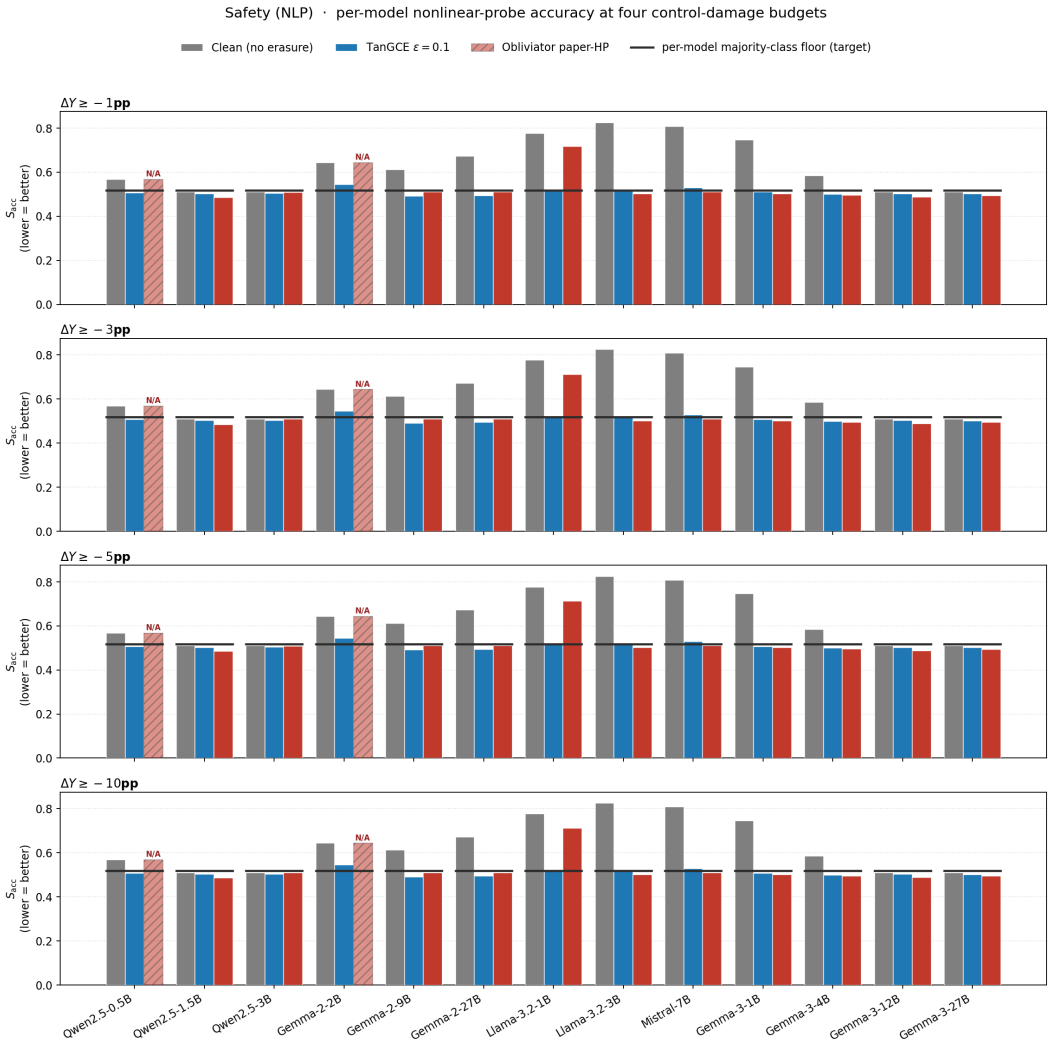


Figure 4: **Safety: per-model nonlinear-probe accuracy at four control-damage budgets.** Layout matches Fig. 2.

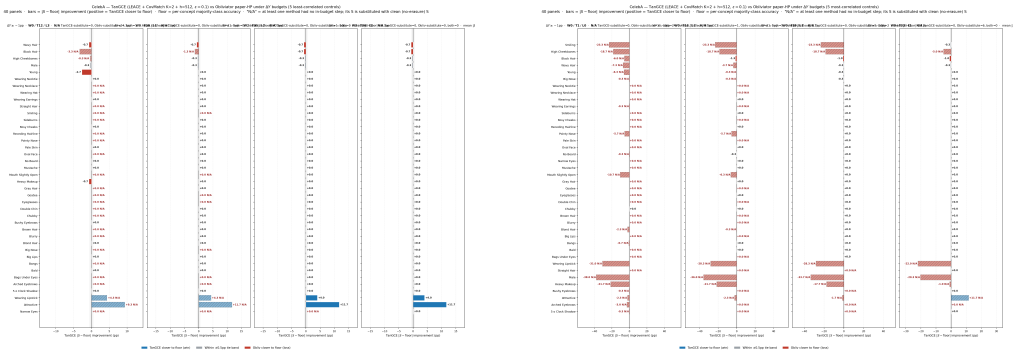
(Eq. 11). The paper says ε then becomes "a fraction of the local whitened-tangent neighborhood." This appendix unpacks what that phrase means in concrete terms.

Reading the trust region as a budget. With $s_i = r_i$ (the radial choice), the bound reads "move at most $\varepsilon\cdot$ Euclidean k-NN radius." This is direction-agnostic: every direction inside ambient space gets the same budget. With s_i defined by Eq. 11 (the FMaHa whitened-tangent radius), the bound instead reads "move at most ε of the neighborhood radius *after normalising each local tangent axis by its own scale*." That distinction is what makes ε direction-aware.

Worked example. Suppose the local tangent space at $\mathbf{x}_i$ is two-dimensional with right singular vectors $\mathbf{v}_1, \mathbf{v}_2$ and singular values $\sigma_{i,1} = 10$, $\sigma_{i,2} = 1$. Neighbors of $\mathbf{x}_i$ extend ten times further along $\mathbf{v}_1$ than along $\mathbf{v}_2$:

$$\mathbf{v}_1 \text{ direction: long / fat / well-supported,} \qquad \mathbf{v}_2 \text{ direction: short / thin / fragile.}$$

The whitened tangent coordinate of a neighbor offset $\mathbf{x}_j - \mathbf{x}_i$ along $\mathbf{v}_k$ is $z_k = \langle \mathbf{x}_j - \mathbf{x}_i, \mathbf{v}_k \rangle / \sigma_{i,k}$. Moving 1 ambient unit along $\mathbf{v}_1$ is therefore $1/10 = 0.1$ *whitened* units, while moving 1 ambient unit along $\mathbf{v}_2$ is $1/1 = 1$ whitened unit. Expressed in whitened coordinates the local neighborhood is approximately spherical: "10% of the whitened neighborhood" is the same allowed move *regardless*



(a) 5 least-correlated controls (low entanglement). (b) 5 most-correlated controls (high entanglement).

Figure 5: **CelebA per-attribute floor-distance win-margin.** Solid blue/red = $\big(|S^{\text{Obliv}} - S_{\text{floor}}| - |S^{TanGCE} - S_{\text{floor}}|\big)\times 100$ on panels where both methods reach the budget; positive (blue) = TanGCE closer to floor. Hatched dark-blue ($\star$) marks walkover panels where TanGCE has at least one in-budget step but Obliviator has none. Tie band ± 0.5pp shaded.

of direction inside the tangent. Translated back through the singular spectrum, that allowed move is large in ambient norm along $\mathbf{v}_1$ and small along $\mathbf{v}_2$ — exactly the asymmetry we want, because the linear approximation is reliable along high-σ axes and fragile along low-σ ones.

Comparison with the Euclidean radius. The Euclidean form is the inequality

$$\|\boldsymbol{\delta}\|_2 \;\leq\; \varepsilon \cdot r_i,$$

which treats every direction equally. The whitened form is approximately

$$\left\|\mathbf{B}_i^\top \boldsymbol{\delta} \,/\, \boldsymbol{\sigma}_i\right\|_2 \;\leq\; \varepsilon \cdot R_i^{\text{white}},$$

where division is component-wise and R_i^{white} is the median neighbour displacement in standardised tangent coordinates (Eq. 11). Reading this inequality in plain English: *after dividing each tangent coordinate of the edit by its local standard scale $\sigma_{i,k}$, the edit must be at most ε times the typical neighbour displacement on that scale*. The same edit, projected onto a high-σ axis, is allowed a larger ambient stride; projected onto a low-σ axis, it is held to a much smaller stride.

Mental model. Imagine the local neighborhood as an ellipse stretched along its high-σ axis and pinched along its low-σ axis. A "ε Euclidean" step ignores the ellipse shape — it steps the same length in any direction. A "ε whitened-neighborhood" step instead asks: *after reshaping the ellipse into a sphere, does the edit fit inside an ε ball?* FMaHa then maps the allowed move back into the original representation space. The result is a direction-aware trust region: it prevents the iterative loop from taking a step that is numerically small in Euclidean norm but huge relative to a thin local tangent direction where the linear approximation is the least reliable.

This is the geometric reason why $\varepsilon = 0.1$ is interpretable as an order-of-magnitude smaller knob than the legacy global λ used in the appendix tables: it bounds the edit relative to the local neighborhood's natural scale on each axis, not to a global step size.

Why ε transfers across datasets without per-panel tuning. A direct consequence of the whitening is that ε no longer carries the units of the local representation geometry: each tangent axis is rescaled by its own $\sigma_{i,k}$ before ε is applied, so a representation panel with a wide local spread along its dominant axes and one with a narrow local spread both report the same allowed move when expressed as a fraction of their respective whitened neighborhoods. This is why we can fix $\varepsilon = 0.1$ once and use it across all 39 NLP panels (13 language models $\times$ 3 concepts at 50%-depth) and all 80 CelebA panels (40 attributes $\times$ 2 control sets) without any per-panel tuning of the trust-region radius. Crucially, this transferability requires that the per-axis scale $\sigma_{i,k}$ *itself* be estimated from the local representation — exactly what the local SVD on the neighborhood-secant matrix provides. The legacy global λ in §4.7 cannot have this property because its units are panel-specific: the same λ moves a Gemma-3-27B representation by a much larger ambient distance than it does a Qwen2.5-0.5B one, which is why the legacy protocol must run a per-panel λ-HPO grid and ours does not.

K CelebA control-set construction (least- vs. most-correlated five attributes)

For each of the 40 binary CelebA attributes used as a target concept, we report two complementary control regimes (Section 5.3, Table 2, Figure 5): the *5 least-correlated* attributes and the *5 most-correlated* attributes, both selected from the remaining 39 attributes. This appendix records exactly how the two control sets are computed.

Computation. Let $\mathbf{Y} \in \{0,1\}^{N_{\text{train}} \times 40}$ be the train-split label matrix, with $\mathbf{Y}_{i,k}$ the label of training example i on attribute k. We compute the empirical Pearson correlation matrix

$$\mathbf{R}_{k\ell} \;=\; \frac{\mathrm{Cov}(\mathbf{Y}_{:,k}, \mathbf{Y}_{:,\ell})}{\sqrt{\mathrm{Var}(\mathbf{Y}_{:,k}) \cdot \mathrm{Var}(\mathbf{Y}_{:,\ell})}} \quad \text{for } k, \ell \in \{1, \ldots, 40\}. \tag{13}$$

For each target attribute k, we then rank the remaining 39 attributes by absolute correlation $|\mathbf{R}_{k\ell}|$:

- the *5 least-correlated* controls are the bottom 5 by $|\mathbf{R}_{k\ell}|$ — they share *minimal* information with the target and isolate clean surgicality;
- the *5 most-correlated* controls are the top 5 by $|\mathbf{R}_{k\ell}|$ — they share *substantial* information with the target and form a stricter test of whether erasure can preserve genuinely related downstream features.

The target attribute itself is excluded from the candidate pool (i.e., $\ell \neq k$). Pairs are computed in float64 with no shrinkage. The full 40×40 correlation matrix and the resulting per-target control sets are stored in `outputs/celeba_correlated_controls.json` (committed for reproducibility) by `scripts/celeba_compute_correlations.py`.

Concrete examples. For *Male*, the 5 most-correlated attributes are *Wearing_Lipstick* ($|r| \approx 0.78$), *No_Beard*, *Heavy_Makeup*, *Attractive*, and *5_o_Clock_Shadow*; the 5 least-correlated are *Narrow_Eyes*, *Blurry*, *Straight_Hair*, *Receding_Hairline*, and *Pale_Skin* (all $|r| < 0.05$). Erasing *Male* while preserving the most-correlated controls is genuinely hard because the controls are partial proxies for the target; erasing while preserving the least-correlated controls tests whether the intervention is geographically narrow but not whether it disentangles correlated information. We therefore report both regimes side by side: each one isolates a different failure mode.

Why two regimes per concept. Reporting only one control set hides whichever regime the method happens to be weak in. A method that destroys the entire representation will pass the least-correlated test (controls are essentially independent of the edit anyway) and fail the most-correlated test catastrophically; conversely, a method that performs no erasure will pass both. The difference between the two regimes per method is therefore informative even when both individual numbers look acceptable. Section 5.3 reads the difference as a measure of how much an erasure method confuses "avoid this concept" with "avoid everything correlated with it".

L Closed-form per-row λ_i statistics

For each panel and each outer iteration of the proposed pipeline (Phase 2 with fixed trust-region $\varepsilon = 0.1$), the closed-form rule of Eq. 10 produces a per-row step magnitude λ_i (capped at $\lambda_{\max} = 64$). Table 19 reports the distribution of these per-row λ_i averaged across all 60 iterations of each trajectory and then aggregated across panels.

The mean across the entire grid, $\bar{\lambda} \approx 29.31$, sits well above the legacy global-λ HPO grid's typical winners ($\lambda^* \in \{1, 2, 4\}$ on most panels for the canonical AmbGCE/TanGCE form). This justifies the apples-to-apples ablation reported in Table 8: when AmbGCE is forced to use a single global $\lambda = \lambda_{\text{TanGCE}} = 29.31$ — i.e., the same effective step magnitude TanGCE's per-row rule produces — its iterative behaviour and final leakage are directly comparable, with the only difference being the gradient-direction constraint (tangent-projected vs. ambient).

M Anisotropy regimes for $\mathbf{C}_i^\alpha$

The σ-preconditioned update of Eq. 12 reweights each tangent direction by $\sigma_{i,k}^{2\alpha}$, where $\alpha \geq 0$ is an anisotropy exponent. Three values of α have natural interpretations:

- $\alpha = 0$: $\mathbf{C}_i^0 = \mathbf{I}$, the update reduces to Eq. 8. Reported as an ablation.
- $\alpha = \frac{1}{2}$: a *mild* anisotropy. Each direction is reweighted by $\sigma_{i,k}$ rather than by $\sigma_{i,k}^2$, preserving information from low-variance directions while still respecting the dominant geometry. **This is the default in this paper.**
- $\alpha = 1$: full Mahalanobis preconditioning. The update direction $\mathbf{C}_i\mathbf{v}_i$ is the first-order optimum of $\min_{\boldsymbol{\delta}} \langle \mathbf{g}_i, \boldsymbol{\delta} \rangle$ subject to the Mahalanobis trust region $\boldsymbol{\delta}^\top \mathbf{C}_i^{-1} \boldsymbol{\delta} \leq \varepsilon^2$. Reported as an ablation; risks suppressing low-variance signal that may carry residual concept information.

Relation to Mahalanobis whitening (and what $\alpha = \frac{1}{2}$ is *not*). A reader familiar with Mahalanobis whitening might wonder whether $\alpha = \frac{1}{2}$ is "the whitening" we are doing. It is not. Mahalanobis whitening transforms the data so the covariance becomes identity, $\mathbf{x}^{\text{white}} = \mathbf{C}^{-1/2}\mathbf{x}$, and corresponds to the *negative* half-power of $\mathbf{C}$ (i.e. $\alpha = -\frac{1}{2}$ in our notation). Whitening makes every direction equal-weighted, which would actively dilute the concept-specific signal we want to suppress and force every tangent axis to receive equal step magnitude.

We use $\mathbf{C}^{+1/2}$ (positive half-power), which does the opposite: it reweights the tangent step *toward* the high-$\sigma_{i,k}$ principal axes — exactly the directions where the local manifold has the most support and where the linear approximation is most reliable — and away from low-σ thin tail directions where the linear approximation is fragile and where motion most easily leaves the manifold. The three values $\alpha \in \{0, \frac{1}{2}, 1\}$ thus form a one-parameter family interpolating between (i) treating every tangent direction as equally trustworthy ($\alpha = 0$, isotropic), (ii) respecting the local ellipsoid mildly with $\sigma_{i,k}$ scaling ($\alpha = \frac{1}{2}$, our default), and (iii) the natural-gradient / Mahalanobis trust-region optimum that maximally suppresses thin directions ($\alpha = 1$). Whitening ($\alpha = -\frac{1}{2}$) sits outside this useful range for our problem because it has the opposite sign of the geometric correction we want; we therefore do not include it.

N Coordinate-chart definition of the tangent space

We adopt the formal definition from Tu [2011]. Fix $\mathbf{x} \in \mathcal{M}$. A small local neighborhood $U \subset \mathcal{M}$ around $\mathbf{x}$ admits a coordinate map $\varphi : U \to \Omega \subset \mathbb{R}^m$ with $\varphi(\mathbf{x}) = 0$. The inverse $\varphi^{-1} : \Omega \to U \subset \mathbb{R}^d$ is a local parametrization back into the ambient hidden space. The tangent space at $\mathbf{x}$ is the m-dimensional linear subspace

$$T_{\mathbf{x}}\mathcal{M} = d(\varphi^{-1})_0(\mathbb{R}^m) \subset \mathbb{R}^d, \tag{14}$$

where $d(\varphi^{-1})_0$ is the differential of φ^{-1} at the origin (its Jacobian, in ML language). It maps an infinitesimal coordinate displacement in $\mathbb{R}^m$ to the corresponding first-order ambient displacement. Because $\mathcal{M}$ is embedded in $\mathbb{R}^d$, $T_{\mathbf{x}}\mathcal{M}$ is an m-dimensional linear subspace of the ambient hidden space.

In TanGCE we never construct φ or φ^{-1}. The leading right singular vectors of the local secant matrix (Eq. 4) serve as an empirical first-order basis for $T_{\mathbf{x}}\mathcal{M}$, and the orthogonal projector $\mathbf{P}_{T_i} = \mathbf{B}_i\mathbf{B}_i^\top$ implements the tangent component $\mathbf{g}^\top$ of any ambient vector $\mathbf{g}$. We do not claim that $\mathcal{M}$ is smooth in any theorem-level sense, only that this approximation tracks the local sample structure well enough to attenuate updates in directions unsupported by it.

O Per-panel breakdown of the headline comparison

Table 20 reports, for every one of the 39 NLP panels and each of the four control-damage budgets used in the main text, the signed deviation from the per-panel majority-class floor for all five methods compared in Tab. 1. Each entry is the trajectory step closest to floor (smallest $|S - S_{\text{floor}}|$) subject to the budget. The trailing *TanGCE W/T/L* row counts panels where TanGCE is strictly closer to floor than the best other method (W), within 0.5pp of it (T), or strictly further (L).

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14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

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16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

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Table 2: **CelebA visual attributes — surgicality under entangled vs. non-entangled controls.** For each of the 40 binary CelebA attributes, we report nonlinear-probe leakage S_{acc} under two control sets: *5 least-correlated* attributes (low entanglement, easier surgicality) and *5 most-correlated* attributes (high entanglement, harder surgicality). S_{acc} is the lowest probe accuracy reached on a trajectory step satisfying mean $\Delta Y \geq -5$pp on the 5 controls. The **Majority** column reports the per-concept majority-class probe-accuracy floor on the test split (the trivial lower bound any informative probe must beat); a method that drives S_{acc} to this value has fully erased the concept under the control budget. Both methods use `paper-HP` hyperparameters from their respective references; TanGCE adds the Phase-1 LEACE + rank-2 covariance projection plus $h = 512$, $\varepsilon = 0.1$. Lower is better (closer to Majority). Bold marks the lower of the two methods per row.

		5 least-correlated controls		5 most-correlated controls	
Concept	Majority	TanGCE	Obliv	TanGCE	Obliv
5_o_Clock_Shadow	0.910	**0.903**	0.910	0.903	–
Arched_Eyebrows	0.733	**0.700**	0.733	0.707	–
Attractive	0.560	0.480	**0.440**	0.697	–
Bags_Under_Eyes	0.777	**0.727**	0.777	**0.763**	0.777
Bald	0.983	0.983	0.983	0.983	0.983
Bangs	0.833	**0.827**	0.833	0.833	0.833
Big_Lips	0.650	**0.623**	0.650	**0.617**	0.650
Big_Nose	0.767	**0.713**	0.767	**0.680**	0.767
Black_Hair	0.703	0.710	**0.703**	0.713	**0.703**
Blond_Hair	0.887	0.887	0.887	0.887	0.887
Blurry	0.943	0.943	0.943	0.943	0.943
Brown_Hair	0.823	**0.773**	0.823	**0.793**	0.823
Bushy_Eyebrows	0.883	**0.857**	0.883	0.873	–
Chubby	0.957	0.957	0.957	0.957	0.957
Double_Chin	0.960	**0.957**	0.960	**0.950**	0.960
Eyeglasses	0.937	**0.930**	0.937	**0.910**	0.937
Goatee	0.957	0.957	0.957	0.957	0.957
Gray_Hair	0.947	0.947	0.947	0.947	0.947
Heavy_Makeup	0.600	**0.533**	0.600	0.777	–
High_Cheekbones	0.503	**0.500**	0.503	–	–
Male	0.617	**0.533**	0.617	0.953	–
Mouth_Slightly_Open	0.523	**0.477**	0.483	**0.477**	0.523
Mustache	0.960	0.960	0.960	0.960	0.960
Narrow_Eyes	0.847	0.847	–	0.847	0.847
No_Beard	0.873	**0.850**	0.873	**0.870**	0.873
Oval_Face	0.693	**0.657**	0.693	**0.660**	0.693
Pale_Skin	0.980	**0.960**	0.980	**0.973**	0.980
Pointy_Nose	0.713	**0.657**	0.713	**0.657**	0.713
Receding_Hairline	0.917	0.917	0.917	**0.913**	0.917
Rosy_Cheeks	0.940	0.940	0.940	0.940	0.940
Sideburns	0.967	0.967	0.967	0.967	0.967
Smiling	0.520	**0.477**	0.520	–	–
Straight_Hair	0.827	**0.807**	0.827	0.807	–
Wavy_Hair	0.633	**0.600**	0.633	**0.607**	0.633
Wearing_Earrings	0.807	**0.777**	0.807	**0.770**	0.807
Wearing_Hat	0.967	**0.957**	0.967	**0.960**	0.967
Wearing_Lipstick	0.523	0.517	**0.477**	0.853	–
Wearing_Necklace	0.857	**0.843**	0.857	0.857	0.857
Wearing_Necktie	0.937	0.937	0.937	0.937	0.937
Young	0.753	**0.710**	0.753	**0.697**	0.753
Mean	0.804	**0.782**	0.798	**0.831**	0.850

Table 3: Full erasure method comparison at layer 8 across three concept domains (Qwen2.5-0.5B), plus a Majority Class reference floor. Each λ-based method's λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy; the same probe architecture (MLP hidden 256, 1500 steps, AdamW lr=10^{-3}) is used for both HPO and final evaluation, ensuring no ranking distortion from mismatched capacity. IGBP and linear baselines use their standard procedure. Majority Class is not an erasure method; it reports the label-imbalance floor implied by the test split. L/NL Leak: retrained linear logistic-regression / nonlinear MLP probe accuracy on the target concept (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change relative to clean representations (closer to zero = more surgical). Control tasks: answer matching (sycophancy), profession classification (gender), helpfulness (safety).

	Sycophancy			Gender (Bias in Bios)			Safety		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.685	.748	.000	.719	.758	.000	.798	.805	+.003
Random Per-Samp.	.621	.745	.000	.723	.743	+.003	.784	.799	−.010
Linear Proj.	.683	.740	**.000**	.722	.747	+.001	.807	.800	**−.003**
INLP	.683	.734	**.000**	.722	.737	+.001	.807	.795	−.004
LEACE	.681	.681	**.000**	.719	.739	−.002	.801	.785	−.015
IGBP	.683	.748	−.011	.660	.675	−.015	.777	.759	+.062
AmbGCE	.714	.703	**.000**	.663	.701	**.000**	.780	.778	−.040
TanGCE	**.628**	**.602**	−.015	**.603**	**.618**	−.008	**.654**	**.663**	+.039

Table 4: CLIP ViT-B/32 concept erasure on CelebA (validation HPO for λ). Each method's λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy; the same probe (MLP hidden 256, 1500 steps, AdamW lr=10^{-3}) is used for both HPO and evaluation. L/NL Leak: retrained linear / nonlinear probe accuracy on test (lower = better). ΔCtrl: mean control-task accuracy change across 5 least-correlated attributes (closer to zero = more surgical). Representation: CLIP ViT-B/32 pooled embedding (768 dimensions).

		Male			Smiling			Blond_Hair		
Method	λ^*	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	0.5	.997	.990	−0.3	.903	.903	−0.3	.937	.923	−0.3
Random Per-Samp.	0.5	.993	.990	−0.1	.867	.887	−1.3	.933	.920	+0.1
Linear Proj.	1.0	.987	.987	−0.7	.903	.913	**+0.1**	.923	.940	−1.4
INLP	1.0	.587	.987	**−0.2**	.580	.740	+0.6	.713	.917	**+0.7**
LEACE	0.5	.997	.997	−0.3	.503	.820	+0.4	.903	.917	−0.1
AmbGCE	0.5	.997	.997	−0.9	.787	.877	**+0.1**	.910	.923	−1.1
TanGCE	**8.0**	**.800**	**.847**	−1.1	**.487**	**.443**	−0.1	**.693**	**.720**	−1.6

Table 9: Per-layer breakdown of TanGCE vs. AmbGCE across models and concepts. Each model is evaluated at three layers corresponding to 25%, 50%, and 75% of its total depth. NL: nonlinear probe accuracy (lower = stronger erasure). ΔCtrl: downstream accuracy change (closer to zero = more surgical). λ^* selected from $\{0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0\}$.

			Sycophancy		Gender		Safety	
Model	Method	Layer	NL↓	ΔCtrl	NL↓	ΔCtrl	NL↓	ΔCtrl
Qwen2.5-0.5B	AmbGCE	6	0.704	+0.000	0.724	+0.001	0.765	−0.043
	TanGCE	6	**0.592**	+0.000	**0.564**	−0.037	**0.669**	+0.022
	AmbGCE	12	0.781	+0.000	0.715	−0.002	0.806	−0.037
	TanGCE	12	**0.724**	+0.000	**0.538**	−0.060	**0.713**	−0.002
	AmbGCE	18	0.933	+0.000	0.870	−0.008	0.847	−0.042
	TanGCE	18	**0.838**	−0.000	**0.634**	−0.017	**0.753**	+0.002
Qwen2.5-1.5B	AmbGCE	7	0.731	+0.000	0.733	−0.002	0.781	−0.010
	TanGCE	7	**0.672**	+0.000	**0.560**	−0.070	**0.691**	+0.021
	AmbGCE	14	0.798	+0.000	0.858	−0.002	0.826	−0.011
	TanGCE	14	**0.675**	−0.007	**0.577**	−0.053	**0.692**	−0.077
	AmbGCE	21	0.907	+0.000	0.935	−0.001	0.865	−0.047
	TanGCE	21	**0.765**	−0.001	**0.658**	−0.005	**0.769**	−0.073
Qwen2.5-3B	AmbGCE	9	0.808	+0.000	0.855	−0.007	0.787	−0.002
	TanGCE	9	**0.752**	−0.011	**0.550**	−0.018	**0.672**	−0.024
	AmbGCE	18	0.821	+0.000	0.870	−0.001	0.823	−0.017
	TanGCE	18	**0.761**	+0.000	**0.585**	−0.040	**0.698**	−0.056
	AmbGCE	27	0.953	+0.000	0.930	+0.002	0.895	−0.002
	TanGCE	27	**0.906**	+0.000	**0.634**	−0.018	**0.839**	−0.095
Gemma-2-2B	AmbGCE	6	0.628	+0.000	0.750	+0.000	0.718	−0.015
	TanGCE	6	**0.603**	+0.000	**0.545**	−0.010	**0.550**	+0.001
	AmbGCE	13	0.668	+0.000	0.795	+0.003	0.760	−0.010
	TanGCE	13	**0.632**	+0.000	**0.532**	−0.000	**0.572**	−0.017
	AmbGCE	20	0.673	+0.000	0.885	+0.003	0.757	−0.019
	TanGCE	20	**0.641**	+0.000	**0.642**	−0.025	**0.610**	−0.004
Gemma-2-9B	AmbGCE	10	0.686	+0.000	0.813	−0.005	0.801	−0.006
	TanGCE	10	**0.684**	−0.000	**0.535**	−0.034	**0.665**	+0.006
	AmbGCE	21	0.770	+0.000	0.768	+0.001	0.751	−0.024
	TanGCE	21	**0.672**	+0.000	**0.664**	−0.015	**0.664**	+0.000
	AmbGCE	32	0.780	+0.000	0.927	−0.009	0.763	−0.034
	TanGCE	32	**0.696**	+0.000	**0.727**	−0.032	**0.664**	−0.002
Gemma-2-27B	AmbGCE	12	0.812	+0.000	0.890	−0.003	0.809	−0.005
	TanGCE	12	**0.680**	+0.000	**0.655**	−0.057	**0.716**	−0.002
	AmbGCE	23	0.764	+0.000	0.819	−0.003	0.818	−0.098
	TanGCE	23	**0.665**	−0.001	**0.659**	−0.028	**0.685**	−0.011
	AmbGCE	34	0.701	+0.000	0.889	−0.000	0.821	−0.014
	TanGCE	34	**0.670**	+0.000	**0.695**	−0.079	**0.725**	−0.006
Llama-3.2-1B	AmbGCE	4	0.790	+0.000	0.856	−0.001	0.790	−0.028
	TanGCE	4	**0.691**	+0.000	**0.565**	−0.030	**0.612**	+0.008
	AmbGCE	8	0.887	+0.000	0.901	+0.003	0.839	−0.039
	TanGCE	8	**0.704**	+0.000	**0.550**	−0.019	**0.660**	−0.017
	AmbGCE	12	0.815	+0.000	0.935	−0.001	0.853	−0.027
	TanGCE	12	**0.734**	+0.000	**0.755**	−0.012	**0.715**	−0.029
Llama-3.2-3B	AmbGCE	7	0.813	+0.000	0.903	−0.002	0.820	−0.016
	TanGCE	7	**0.728**	−0.001	**0.571**	−0.017	**0.649**	−0.007
	AmbGCE	14	0.909	+0.000	0.929	−0.001	0.860	−0.046
	TanGCE	14	**0.768**	+0.000	**0.610**	−0.010	**0.745**	−0.023
	AmbGCE	21	0.924	+0.000	0.945	−0.005	0.872	−0.027
	TanGCE	21	**0.747**	+0.000	**0.758**	−0.009	**0.733**	−0.030
Mistral-7B	AmbGCE	8	0.820	+0.000	0.895	−0.004	0.852	−0.020
	TanGCE	8	**0.730**	+0.000	**0.599**	−0.012	**0.692**	−0.005
	AmbGCE	16	0.935	+0.000	0.914	−0.003	0.884	−0.032
	TanGCE	16	**0.724**	+0.000	**0.596**	−0.028	**0.730**	−0.038
	AmbGCE	24	0.938	+0.000	0.926	−0.002	0.879	−0.039
	TanGCE	24	**0.721**	−0.001	**0.708**	−0.009	**0.748**	−0.018

Table 10: λ sweep comparing AmbGCE and TanGCE at fixed step sizes across three concept domains (Qwen2.5-0.5B, layer 8). NL: retrained nonlinear MLP probe accuracy (hidden 256, 1500 steps, AdamW lr=10^{-3}; same architecture used for HPO and evaluation) on the target concept (lower = stronger erasure). Both methods use the same GCE concept scorer (hidden 256, 1500 steps). AmbGCE becomes less effective at high λ under the current protocol, while TanGCE continues improving. We interpret this as supportive evidence for the local manifold motivation of tangent-constrained updates, not as a complete characterization of the leakage-utility frontier. The expanded HPO grid $\{0.25, \ldots, 32.0\}$ used in the multi-model experiments further tests this range. Bold: best NL per domain.

		NL↓ at $\lambda =$		
Concept	Method	0.5	4.0	8.0
Sycophancy	AmbGCE	.678	.694	.734
	TanGCE	.662	.596	**.588**
Gender	AmbGCE	.666	.700	.702
	TanGCE	.704	.580	**.470**
Safety	AmbGCE	.664	.776	.770
	TanGCE	.748	.680	**.646**

Table 11: **Phase-1 closed-form ablations on the 39-panel NLP grid (50%-depth layer).** Same fresh nonlinear probe ($h = 128$, 200 SGD steps, patience 3) is trained on each post-erasure representation. *LEACE* is the affine LEACE eraser. *LEACE + CovMatch K=2* additionally projects out the top-2 eigenvectors of $\Delta\Sigma = \Sigma_+ - \Sigma_-$ (rank-3 closed-form erasure total). *TanGCE* is the proposed full pipeline (Phase 1 + iterative tangent loop, ε=0.1); we report the best $|S-\text{floor}|$ trajectory step subject to $\Delta Y \geq -5$pp. NL = nonlinear probe accuracy on test (lower is better; floor is the per-panel majority class). ΔY in pp on the control task. Bold marks the row's NL minimum.

Model	Concept	Floor (NL*)	LEACE NL	LEACE ΔYpp	LEACE + CovMatch K=2 NL	LEACE + CovMatch K=2 ΔYpp	TanGCE ε=0.1 NL	TanGCE ε=0.1 ΔYpp
Qwen2.5-0.5B	Sycophancy	0.550	**0.550**	+0.0	**0.550**	-2.1	**0.550**	-0.1
	Gender	0.539	0.709	-0.1	0.684	-0.4	**0.549**	-4.1
	Safety	0.510	0.752	+0.0	0.529	+0.0	**0.510**	-0.1
Qwen2.5-1.5B	Sycophancy	0.550	**0.550**	-0.1	**0.550**	-1.5	**0.550**	+1.2
	Gender	0.539	0.822	-0.3	0.788	-1.7	**0.592**	-3.9
	Safety	0.510	0.759	+2.9	**0.510**	+2.9	**0.510**	+0.0
Qwen2.5-3B	Sycophancy	0.675	**0.675**	+0.1	**0.675**	-2.7	**0.675**	-0.7
	Gender	0.539	0.857	-1.3	0.783	-0.1	**0.548**	-4.1
	Safety	0.510	0.718	+0.5	0.520	+0.5	**0.510**	+0.0
Gemma-2-2B	Sycophancy	0.543	0.643	+0.0	**0.543**	-0.4	**0.543**	-0.2
	Gender	0.539	0.722	+0.3	0.681	-0.7	**0.539**	-4.9
	Safety	0.510	0.715	+0.7	0.654	+0.7	**0.510**	+0.0
Gemma-2-9B	Sycophancy	0.569	0.751	-0.1	**0.569**	-0.7	0.569	+0.0
	Gender	0.539	0.638	+0.2	0.562	-0.4	**0.539**	-3.5
	Safety	0.510	0.623	+0.0	**0.510**	+0.0	**0.510**	+0.0
Gemma-2-27B	Sycophancy	0.553	0.695	-0.2	**0.553**	-0.9	**0.553**	-0.3
	Gender	0.539	0.663	+0.0	0.605	-2.5	**0.539**	-0.5
	Safety	0.510	0.736	+0.7	0.643	+1.4	**0.510**	+0.0
Llama-3.2-1B	Sycophancy	0.551	0.849	-0.1	**0.551**	-0.2	**0.551**	+0.0
	Gender	0.539	0.855	-0.4	0.738	+0.8	**0.536**	-3.9
	Safety	0.510	0.817	+0.3	0.778	+0.3	**0.498**	+0.0
Llama-3.2-3B	Sycophancy	0.551	0.896	-0.0	**0.551**	+0.0	**0.551**	+0.0
	Gender	0.539	0.889	+0.2	0.794	-0.4	**0.535**	-2.8
	Safety	0.510	0.837	+0.1	0.821	+0.2	**0.526**	+0.0
Mistral-7B	Sycophancy	0.551	0.920	+0.0	**0.551**	-0.1	**0.551**	+0.2
	Gender	0.539	0.851	-0.4	0.815	-0.3	**0.606**	-4.8
	Safety	0.510	0.825	+0.0	0.811	+0.0	**0.532**	+0.2
Gemma-3-1B	Sycophancy	0.551	0.727	-0.1	**0.551**	-2.2	**0.551**	-0.0
	Gender	0.539	0.708	+0.0	0.613	-0.8	**0.539**	+2.0
	Safety	0.510	0.726	+0.0	0.723	-1.0	**0.510**	+1.0
Gemma-3-4B	Sycophancy	0.551	**0.551**	-0.0	**0.551**	-0.9	**0.551**	-0.4
	Gender	0.539	0.704	-0.5	0.580	-2.7	**0.539**	+4.9
	Safety	0.510	0.735	+1.0	0.609	+1.0	**0.510**	-0.2
Gemma-3-12B	Sycophancy	0.556	0.613	+0.0	**0.556**	-0.5	**0.556**	-0.5
	Gender	0.539	**0.539**	+3.5	0.606	-8.7	0.539	+3.1
	Safety	0.510	0.589	+3.8	**0.510**	+3.8	**0.510**	+0.0
Gemma-3-27B	Sycophancy	0.526	0.586	-0.2	0.566	-0.1	**0.526**	+0.0
	Gender	0.539	0.700	+1.7	0.678	-2.6	**0.539**	-4.8
	Safety	0.510	0.686	+0.9	**0.510**	+0.7	**0.510**	+0.0
Mean sycophancy		0.560	0.693	-0.0	0.563	-0.9	**0.560**	-0.1
Mean gender		0.539	0.743	+0.2	0.687	-1.6	**0.549**	-2.1
Mean safety		0.510	0.732	+0.8	0.625	+0.8	**0.512**	+0.1

Table 12: CelebA all-40-attribute summary (CLIP ViT-B/32). The upper block reports methods evaluated under the validation-HPO protocol from the language-model experiments: λ^* is selected per method per concept from the grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL probe accuracy. The lower block reports trajectory-based methods (TanGCE+LEACE preprocessing and Obliviator) for which we instead pick the trajectory step with lowest probe accuracy subject to mean $\Delta Y \geq -5$pp on the 5 control attributes. NL Leak: mean nonlinear test accuracy across 40 concepts (lower = better). Mean ΔCtrl: mean control accuracy change averaged over 40 concepts. Worst ΔCtrl: worst single control drop averaged over 40 concepts (only available for λ-HPO methods that report per-control values).

Method	NL Leak↓	Mean ΔCtrl	Worst ΔCtrl	Med. λ^*
λ-HPO selection ($\lambda^* \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$, lowest validation NL):				
Random Global	0.883	-0.4%	-1.6%	2.0
Random Per-Samp.	0.882	-0.3%	$\mathbf{-1.2\%}$	0.5
Linear Proj.	0.869	-0.8%	-2.7%	4.0
INLP	0.859	-3.5%	-6.9%	1.0
LEACE	0.839	$\mathbf{-0.2\%}$	$\mathbf{-1.2\%}$	2.0
AmbGCE	0.875	-0.6%	-2.8%	0.5
TanGCE (canonical, $\alpha = \frac{1}{2}$)	0.707	-0.7%	-2.1%	8.0
Trajectory-based selection (best step at $\Delta Y \geq -5$pp):				
Obliviator	0.799	-2.1%	—	—
TanGCE+LEACE+CovMatch K=2 (ours)	**0.777**	$\mathbf{-1.1\%}$	—	—

Table 13: Supplementary deeper-layer sycophancy comparison at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is answer-matching change relative to clean representations. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table is included as supplementary evidence of directional consistency with the layer-8 comparison in Appendix Table 3. Deeper-layer tangent runs continue to reduce leakage more than ambient baselines across this sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.063	0.213	-20.7pp	-0.5pp	2.88	0.62
16	0.153	0.400	+0.0pp	+0.0pp	3.79	0.27
23	0.138	0.361	-0.1pp	-0.2pp	44.50	3.11

Table 14: Supplementary deeper-layer safety comparison at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is pairwise helpfulness change. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table is included as supplementary evidence of directional consistency with the layer-8 comparison in Appendix Table 3. Deeper-layer tangent runs continue to reduce leakage more than ambient baselines across this sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.0725	0.2583	-0.052	-0.023	2.70	0.18
16	0.1292	0.3517	-0.092	+0.003	4.37	0.60
23	0.1483	0.3404	-0.062	-0.152	38.32	5.30

Table 15: Local Table-2-style CelebA visual comparison using CLIP ViT-B/32 embeddings. For each target concept we intervene once per method and evaluate (i) concept leakage via retrained linear/nonlinear probes (lower is better) and (ii) mean control-task accuracy change across 3 least-correlated control attributes (closer to zero is better). Targets and controls: Male $\rightarrow$ Blurry,Narrow_Eyes,Straight_Hair; Smiling $\rightarrow$ Straight_Hair,Bushy_Eyebrows,Gray_Hair; Blond_Hair $\rightarrow$ Straight_Hair,Narrow_Eyes,Blurry. Lambdas follow the layer-8 single-model protocol (Appendix Table 3): all methods λ=0.5 except TanGCE λ=8.0.

	Male			Smiling			Blond_Hair		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.993	.994	-0.0pp	.891	.911	+0.0pp	.909	.933	-0.1pp
Random Per-Samp.	.993	.994	+0.0pp	.891	.907	+0.2pp	.907	.932	-0.1pp
Linear Proj.	.993	.992	+0.0pp	.900	.906	-0.3pp	.919	.931	-0.2pp
INLP	.994	.993	-0.0pp	.900	.903	+0.3pp	.918	.932	-0.0pp
LEACE	.994	.991	-0.0pp	.894	.894	+0.2pp	.909	.925	-0.1pp
AmbGCE	.990	.995	-0.2pp	.858	.884	-0.4pp	.893	.920	+0.2pp
TanGCE	.887	.953	-0.8pp	.649	.649	-0.6pp	.699	.747	-2.1pp

Table 16: Per-concept nonlinear probe leakage on CelebA (CLIP ViT-B/32), all 40 binary attributes. Each cell is retrained nonlinear probe test accuracy after erasure (lower = stronger erasure). λ^* is selected per method per concept from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL accuracy. Bold marks the lowest leakage per row. Abbrev.: RandG = Random Global, RandS = Random Per-Sample, LinProj = Linear Projection, AmbGCE = ambient gradient-based concept erasure, TanGCE = tangent gradient-based concept erasure.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	.92	.92	.91	.92	.87	.92	.91	**.77**
Arched_Eyebrows	.79	.79	.76	.76	.69	.77	.77	**.59**
Attractive	.73	.74	.69	.73	.63	**.53**	.73	.60
Bags_Under_Eyes	.76	.75	.74	.75	.70	.77	.75	**.69**
Bald	1.00	.99	.99	.99	.98	1.00	.99	**.97**
Bangs	.91	.91	.89	.89	.87	.92	.91	**.68**
Big_Lips	.80	.79	.74	.80	**.68**	.82	.81	.73
Big_Nose	.76	.76	.73	.74	.72	.73	.76	**.61**
Black_Hair	.85	.84	.82	.81	.79	.83	.82	**.46**
Blond_Hair	.92	.92	.94	.92	.92	.92	.92	**.72**
Blurry	.98	.98	.98	.96	.97	.98	.97	**.88**
Brown_Hair	.80	.80	.76	.75	.75	.80	.80	**.52**
Bushy_Eyebrows	.87	.88	.86	.85	.76	.87	.86	**.66**
Chubby	.94	.94	.95	.95	.92	.95	.94	**.81**
Double_Chin	.94	.94	.94	.93	.92	.94	.95	**.85**
Eyeglasses	.99	1.00	.96	.93	.96	1.00	.96	**.90**
Goatee	.96	.97	.97	.93	.96	.96	.95	**.85**
Gray_Hair	.97	.97	.97	.97	.97	.97	.98	**.89**
Heavy_Makeup	.89	.89	.89	.86	.85	.88	.88	**.70**
High_Cheekbones	.79	.79	.77	.75	.70	.57	.76	**.50**
Male	.99	.99	.99	.99	1.00	.98	1.00	**.85**
Mouth_Slightly_Open	.89	.89	.89	.71	.72	.66	.80	**.43**
Mustache	.94	.95	.95	.94	.92	.95	.93	**.77**
Narrow_Eyes	.88	.88	.80	.87	.80	.88	.88	**.78**
No_Beard	.94	.95	.94	.93	.92	.94	.95	**.77**
Oval_Face	.70	.71	.69	.68	**.67**	.69	.67	.72
Pale_Skin	.93	.93	.94	.95	.93	.92	.93	**.83**
Pointy_Nose	.75	.72	.71	.69	**.63**	.72	.75	.64
Receding_Hairline	.94	.94	.96	.96	.94	.94	.95	**.69**
Rosy_Cheeks	.93	.93	.93	.94	.91	.94	.93	**.79**
Sideburns	.95	.95	.94	.94	.95	.94	.94	**.82**
Smiling	.90	.89	.91	.74	.82	.59	.88	**.44**
Straight_Hair	.78	.78	.79	.77	.78	.78	.78	**.49**
Wavy_Hair	.78	.76	.77	.77	.78	.77	.78	**.52**
Wearing_Earrings	.90	.89	.86	.88	.85	.89	.88	**.61**
Wearing_Hat	1.00	1.00	.97	.97	.97	1.00	.97	**.94**
Wearing_Lipstick	.90	.90	.90	.89	.85	.88	.92	**.75**
Wearing_Necklace	.84	.85	.80	.80	.77	.85	.83	**.68**
Wearing_Necktie	.93	.93	.92	.92	.91	.92	.93	**.78**
Young	.86	.86	.85	.84	.83	.86	.87	**.60**
Mean	.88	.88	.87	.86	.84	.86	.87	**.71**

Table 17: Per-concept mean control accuracy change ΔCtrl (%) on CelebA, all 40 binary attributes. For each concept, ΔCtrl is averaged over the 5 least-correlated control attributes (closer to zero = more surgical). λ^* selection matches Table 16.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	+0.0	+0.1	-0.7	+0.1	+0.0	+0.1	-2.3	-0.2
Arched_Eyebrows	-1.3	-2.3	-1.2	-0.8	-1.1	-2.5	+0.6	-0.6
Attractive	-0.3	-1.0	-1.7	-0.8	-0.3	-0.9	+0.0	-0.8
Bags_Under_Eyes	-3.7	-1.7	-0.3	-7.7	-0.3	-0.3	-0.3	-0.1
Bald	+0.1	+0.0	-0.1	+0.1	-0.1	+1.1	+2.3	+0.2
Bangs	-3.7	-1.2	-0.5	-7.9	-0.1	-0.7	-0.3	-0.8
Big_Lips	+0.1	-0.0	-0.3	+0.0	-0.1	-0.5	+0.1	-0.1
Big_Nose	-0.1	+0.3	-3.5	-9.7	+0.0	-0.3	-1.8	+0.1
Black_Hair	-0.1	-0.0	+0.1	+0.2	+0.3	-0.3	-0.3	-0.7
Blond_Hair	-0.3	+0.1	-1.4	+0.7	-0.1	-0.4	-1.1	-1.6
Blurry	-1.3	+0.0	-0.1	-5.5	+0.1	-0.7	+0.7	-0.1
Brown_Hair	+0.1	-0.9	-0.3	-6.0	-0.7	+0.4	+0.1	-0.2
Bushy_Eyebrows	+0.0	+0.1	-0.1	-1.5	+0.2	-0.9	-0.3	-0.5
Chubby	-0.1	+0.3	-0.2	-1.4	-0.2	-0.2	+0.4	-2.0
Double_Chin	+0.1	+0.1	-0.4	-12.7	+0.3	+0.7	-0.7	-2.5
Eyeglasses	-0.1	+0.1	-0.0	-0.1	-0.2	-0.4	-0.2	+0.2
Goatee	+0.1	-0.1	+0.0	+0.2	+0.1	+0.5	-0.3	-0.4
Gray_Hair	-0.2	-0.3	-2.1	+0.4	-0.1	-1.1	-1.5	-1.5
Heavy_Makeup	-0.3	-0.1	-1.1	-1.7	-1.3	-1.3	-1.1	-0.9
High_Cheekbones	+0.1	+0.2	+0.1	-0.3	-0.1	-0.4	-0.0	-0.7
Male	-0.3	-0.1	-0.7	-0.2	-0.3	-2.2	-0.9	-1.1
Mouth_Slightly_Open	-0.3	-0.6	+0.1	+0.4	+0.2	+0.3	+0.1	-0.2
Mustache	+0.0	-0.1	+0.1	-10.3	-0.5	+1.0	+0.5	-1.5
Narrow_Eyes	-0.1	-0.2	-0.3	+0.2	-0.1	+0.3	+0.0	+0.0
No_Beard	-1.2	-0.2	-4.0	-6.5	-1.6	-0.1	-0.7	-0.5
Oval_Face	-2.0	+0.1	-0.1	-14.4	-0.6	-0.3	+0.2	-0.3
Pale_Skin	-1.2	-0.1	-0.1	-6.8	-0.9	-0.1	+0.7	-1.0
Pointy_Nose	-0.1	-0.2	+0.3	-1.1	+0.2	-0.7	-0.1	-0.1
Receding_Hairline	-0.1	-0.1	-0.1	-5.9	+0.0	-0.3	+0.1	-1.1
Rosy_Cheeks	-0.5	+0.1	-0.6	-0.1	-0.2	-0.1	-0.1	-1.3
Sideburns	-0.4	+0.0	-0.1	+0.1	+0.1	-0.6	-2.5	-0.1
Smiling	-0.3	-1.3	+0.1	+0.6	+0.4	+0.1	+0.1	-0.1
Straight_Hair	-0.1	-0.2	-0.1	-0.9	-0.1	-0.8	+0.5	-0.7
Wavy_Hair	+0.1	-0.7	-0.9	+0.2	+0.1	-0.4	-0.1	-0.7
Wearing_Earrings	+0.8	-0.3	-1.6	-0.3	-0.1	-1.3	-0.5	-0.3
Wearing_Hat	-0.1	+0.0	-0.1	-0.6	+0.0	+1.0	+0.1	-0.2
Wearing_Lipstick	+0.2	-0.1	-6.1	-21.5	-1.9	-1.2	-1.4	-0.9
Wearing_Necklace	+0.1	-1.7	-1.7	-9.3	+0.3	-1.8	+0.2	-1.2
Wearing_Necktie	+0.0	+0.1	+0.0	+0.2	-0.2	-0.7	-13.4	-1.3
Young	-0.0	-0.7	-1.7	-10.7	+0.1	-0.9	-2.2	-0.8
Mean	-0.4	-0.3	-0.8	-3.5	-0.2	-0.4	-0.6	-0.7

Table 18: Geometry evidence map organizing the appendix diagnostics into four research blocks. Each block addresses a distinct aspect of manifold structure: subspace sufficiency examines whether task-relevant computation concentrates in a low-dimensional linear subspace, perturbation sensitivity tests whether on-manifold directions matter more than orthogonal directions, representation connectivity evaluates whether the manifold remains connected across domains, and multi-layer accumulation tracks error compounding through the network. These diagnostics support the mechanism discussion and motivate the need for manifold-aware erasure methods but do not by themselves establish causality.

Experiment Block	Research Question	Primary Evidence
Subspace sufficiency	Subspace concentration	Top-1 agreement under intrinsic projection
Perturbation sensitivity	High-variance vs. orthogonal directions	KL and top-1 under on/off-subspace noise
Representation connectivity	Shared vs. domain manifolds	Cross-domain interpolation smoothness
Multi-layer accumulation	Error compounding across the network	Simultaneous intrinsic projection across layers

Table 19: **Per-row λ_i statistics for the proposed pipeline ($\varepsilon = 0.1$, $\lambda_{\max} = 64$) on the 39-panel NLP grid at 50%-depth.** Each cell averages the per-row λ_i first across all 60 outer iterations of one trajectory, then across the 13 model panels of the concept. The within-panel p_{10}/p_{90} spread (mean across panels) is roughly $14/45$, i.e. a $\sim 10\times$ ratio between the least- and most-confident rows in a single iteration. The safety cap $\lambda_{\max} = 64$ is observed only sporadically (max 57.32 across 39×60 = 2340 iterations); the closed-form rule rarely binds against it.

Concept	n	mean λ_i	median	min	max
Sycophancy	13	29.58	37.15	4.89	49.38
Gender	13	25.28	23.76	0.45	45.08
Safety	13	33.07	33.30	1.71	57.32
Overall	**39**	**29.31**	**30.19**	**0.45**	**57.32**

Table 20: **Per-panel breakdown.** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-panel majority-class floor, subject to $|\Delta Y| \leq B$pp control-damage budget. Bold = best (smallest absolute deviation from floor) per panel per budget block. '–' = no in-budget step at that budget for that method. Final rows: per-budget per-method mean across the 39 panels and TanGCE W/T/L counts vs. the best other method (0.5pp tie band).

Panel	$\Delta Y \leq 1$pp					$\Delta Y \leq 3$pp					$\Delta Y \leq 5$pp					$\Delta Y \leq 10$pp				
	Tan	Obl	Amb	L+C	LE	Tan	Obl	Amb	L+C	LE	Tan	Obl	Amb	L+C	LE	Tan	Obl	Amb	L+C	LE
Gemma2-27B/Gen L23	**+0.0**	**+0.0**	–	–	+12.3	**+0.0**	**+0.0**	–	+6.6	+12.3	**+0.0**	**+0.0**	–	+6.6	+12.3	**+0.0**	**+0.0**	–	+6.6	+12.3
Gemma2-27B/Saf L23	**-0.1**	-0.7	-1.7	+13.3	+22.5	**-0.1**	-0.7	-1.0	+13.3	+22.5	-0.1	-0.7	**-0.1**	+13.3	+22.5	-0.1	-0.7	**-0.1**	+13.3	+22.5
Gemma2-27B/Syc L23	+0.5	+8.5	+18.7	**+0.0**	+14.3	+0.5	+0.5	+18.7	**+0.0**	+14.3	**-0.0**	+0.5	+15.5	+0.0	+14.3	**-0.0**	+0.5	+15.5	+0.0	+14.3
Gemma2-2B/Gen L13	**+6.0**	+24.0	+18.5	+14.2	+18.3	**+3.4**	+21.8	+18.5	+14.2	+18.3	**+0.0**	+21.8	+18.5	+14.2	+18.3	**+0.0**	+21.8	+18.5	+14.2	+18.3
Gemma2-2B/Saf L13	+2.8	-0.6	**+0.1**	+14.3	+20.5	+2.8	-0.6	**+0.1**	+14.3	+20.5	+2.8	-0.6	**+0.1**	+14.3	+20.5	+2.8	-0.6	**-0.0**	+14.3	+20.5
Gemma2-2B/Syc L13	+0.9	+1.6	+13.0	**+0.0**	+9.9	+0.2	+1.6	+13.0	**+0.0**	+9.9	+0.2	+1.6	+13.0	**+0.0**	+9.9	+0.2	+1.6	+12.9	**+0.0**	+9.9
Gemma2-9B/Gen L21	**+0.0**	+17.4	+15.4	+2.3	+9.9	**+0.0**	+17.4	+15.4	+2.3	+9.9	**+0.0**	+17.4	+15.4	+2.3	+9.9	**+0.0**	+4.8	+15.4	+2.3	+9.9
Gemma2-9B/Saf L21	+0.0	-0.4	+2.0	**+0.0**	+11.3	+0.0	-0.4	+0.1	**+0.0**	+11.3	+0.0	-0.4	+0.1	**+0.0**	+11.3	+0.0	-0.4	+0.1	**+0.0**	+11.3
Gemma2-9B/Syc L21	+1.0	+1.4	+22.1	**-0.0**	+18.2	+0.2	+1.4	+22.1	**-0.0**	+18.2	+0.2	+1.4	+22.1	**-0.0**	+18.2	+0.2	+1.4	+20.9	**-0.0**	+18.2
Gemma3-12B/Gen L24	**+0.0**	**+0.0**	+8.8	–	-0.0	**+0.0**	**+0.0**	-0.9	–	-0.0	**+0.0**	**+0.0**	**+0.0**	–	-0.0	**+0.0**	**+0.0**	**+0.0**	+6.7	-0.0
Gemma3-12B/Saf L24	+0.1	-0.7	+0.1	**+0.0**	+7.9	+0.1	-0.7	+0.1	**+0.0**	+7.9	+0.1	-0.7	+0.1	**+0.0**	+7.9	+0.1	-0.7	+0.1	**+0.0**	+7.9
Gemma3-12B/Syc L24	+0.4	+1.1	–	**+0.0**	+5.7	+0.1	+1.1	+7.7	**+0.0**	+5.7	+0.0	+1.1	+7.7	**+0.0**	+5.7	+0.0	+1.1	+7.7	**+0.0**	+5.7
Gemma3-1B/Gen L13	**+0.0**	+5.0	–	+7.4	+16.9	**+0.0**	+5.0	+15.4	+7.4	+16.9	**+0.0**	+5.0	+15.4	+7.4	+16.9	**+0.0**	**+0.0**	+15.4	+7.4	+16.9
Gemma3-1B/Saf L13	**+0.1**	-0.8	-0.4	+21.2	+21.5	**+0.1**	-0.8	-0.4	+21.2	+21.5	**+0.1**	-0.8	-0.4	+21.2	+21.5	**+0.1**	-0.8	-0.3	+21.2	+21.5
Gemma3-1B/Syc L13	**+0.2**	+0.3	+0.3	–	+17.6	-0.1	+0.3	+0.3	**+0.0**	+17.6	-0.1	+0.3	+0.3	**+0.0**	+17.6	-0.1	+0.3	+0.3	**+0.0**	+17.6
Gemma3-27B/Gen L31	**+0.0**	+15.9	–	–	+16.0	**+0.0**	+15.9	–	+13.9	+16.0	**+0.0**	+15.9	–	+13.9	+16.0	**+0.0**	**+0.0**	–	+13.9	+16.0
Gemma3-27B/Saf L31	-0.2	-0.3	-0.0	**+0.0**	+17.6	-0.2	-0.3	-0.0	**+0.0**	+17.6	-0.2	-0.3	-0.0	**+0.0**	+17.6	-0.2	-0.3	-0.0	**+0.0**	+17.6
Gemma3-27B/Syc L31	**+0.3**	+1.8	+4.3	+4.0	+6.0	**+0.2**	+1.8	+4.3	+4.0	+6.0	**+0.2**	+1.8	+4.3	+4.0	+6.0	**+0.2**	+1.8	+3.5	+4.0	+6.0
Gemma3-4B/Gen L17	**+0.0**	+7.9	–	–	+16.5	**+0.0**	+7.9	+15.0	+4.0	+16.5	**+0.0**	+7.9	+15.0	+4.0	+16.5	**+0.0**	**+0.0**	+15.0	+4.0	+16.5
Gemma3-4B/Saf L17	**-0.0**	-0.8	-0.1	+9.9	+22.5	**-0.0**	-0.8	-0.1	+9.9	+22.5	**-0.0**	-0.8	-0.1	+9.9	+22.5	**-0.0**	-0.8	-0.1	+9.9	+22.5
Gemma3-4B/Syc L17	+0.3	+0.2	–	**+0.0**	**+0.0**	+0.3	+0.2	+0.4	**+0.0**	**+0.0**	-0.0	+0.2	+0.4	**+0.0**	**+0.0**	-0.0	+0.2	+0.4	**+0.0**	**+0.0**
Llama3.2-1B/Gen L8	+16.2	**+15.4**	–	+19.9	+31.6	**+5.8**	+15.4	–	+19.9	+31.6	**+0.0**	+15.4	+25.7	+19.9	+31.6	**+0.0**	+15.4	+25.7	+19.9	+31.6
Llama3.2-1B/Saf L8	**+0.1**	-0.7	-5.7	+26.7	+30.7	**+0.1**	-0.7	-5.7	+26.7	+30.7	**+0.1**	-0.7	-5.7	+26.7	+30.7	**+0.1**	-0.7	-1.7	+26.7	+30.7
Llama3.2-1B/Syc L8	-0.0	+0.3	–	**+0.0**	+29.8	-0.0	+0.3	–	**+0.0**	+29.8	-0.0	+0.3	–	**+0.0**	+29.8	-0.0	+0.3	–	**+0.0**	+29.8
Llama3.2-3B/Gen L14	**+15.1**	+15.8	–	+25.5	+35.0	**-0.1**	+15.8	+30.3	+25.5	+35.0	**-0.1**	+15.8	+30.3	+25.5	+35.0	**+0.0**	+15.8	+30.3	+25.5	+35.0
Llama3.2-3B/Saf L14	**-0.1**	-0.8	-0.7	+31.0	+32.7	**-0.1**	-0.8	-0.7	+31.0	+32.7	**-0.1**	-0.8	-0.7	+31.0	+32.7	**-0.1**	-0.8	-0.7	+31.0	+32.7
Llama3.2-3B/Syc L14	+0.3	+0.3	–	**+0.0**	+34.5	+0.3	+0.3	–	**+0.0**	+34.5	+0.3	+0.3	–	**+0.0**	+34.5	+0.3	+0.3	+0.3	**+0.0**	+34.5
Mistral-7B/Gen L16	**+19.5**	+24.6	–	+27.6	+31.2	**+10.6**	+24.6	+28.6	+27.6	+31.2	**+0.3**	+24.6	+28.6	+27.6	+31.2	**+0.0**	+24.6	+28.6	+27.6	+31.2
Mistral-7B/Saf L16	+1.1	**-0.8**	-2.9	+30.0	+31.5	+1.1	**-0.8**	-2.9	+30.0	+31.5	+1.1	**-0.8**	-2.9	+30.0	+31.5	+1.1	**-0.8**	-2.9	+30.0	+31.5
Mistral-7B/Syc L16	+0.3	+0.3	+36.9	**+0.0**	+37.0	+0.3	+0.3	+36.7	**+0.0**	+37.0	+0.3	+0.3	+36.7	**+0.0**	+37.0	+0.3	+0.3	+36.7	**+0.0**	+37.0
Qwen2.5-0.5B/Gen L12	+8.8	**+6.6**	+15.8	+14.5	+17.0	**+5.9**	+6.6	+15.8	+14.5	+17.0	**+3.0**	+6.6	+15.8	+14.5	+17.0	+1.8	**+0.0**	+15.8	+14.5	+17.0
Qwen2.5-0.5B/Saf L12	**+0.0**	-0.8	+0.1	+1.9	+24.1	**+0.0**	-0.8	+0.1	+1.9	+24.1	**+0.0**	-0.8	+0.1	+1.9	+24.1	**+0.0**	-0.8	+0.1	+1.9	+24.1
Qwen2.5-0.5B/Syc L12	+0.2	+0.3	–	–	**+0.0**	+0.2	+0.3	–	**+0.0**	**+0.0**	+0.2	+0.3	–	**+0.0**	**+0.0**	+0.2	+0.3	–	**+0.0**	**+0.0**
Qwen2.5-1.5B/Gen L14	+13.5	**+13.5**	+28.2	–	+28.2	+11.1	**+8.9**	+28.2	+24.9	+28.2	**+6.8**	+8.9	+28.2	+24.9	+28.2	**+3.9**	+8.9	+28.2	+24.9	+28.2
Qwen2.5-1.5B/Saf L14	-0.0	-0.3	+2.7	**+0.0**	+24.9	-0.0	-0.3	+2.0	**+0.0**	+24.9	-0.0	-0.3	+2.0	**+0.0**	+24.9	-0.0	-0.3	+2.0	**+0.0**	+24.9
Qwen2.5-1.5B/Syc L14	+0.3	+0.3	–	–	**+0.0**	+0.3	+0.3	–	**+0.0**	**+0.0**	+0.3	+0.3	–	**+0.0**	**+0.0**	+0.3	+0.3	–	**+0.0**	**+0.0**
Qwen2.5-3B/Gen L18	**+12.6**	+14.3	–	+24.4	–	**+8.1**	+14.3	+26.5	+24.4	+31.8	**+7.1**	+14.3	+26.5	+24.4	+31.8	**+3.8**	+14.3	+26.5	+24.4	+31.8
Qwen2.5-3B/Saf L18	**+0.0**	-0.3	-0.2	+0.9	+20.8	**+0.0**	-0.3	**+0.0**	+0.9	+20.8	**+0.0**	-0.3	**+0.0**	+0.9	+20.8	**+0.0**	-0.3	**+0.0**	+0.9	+20.8
Qwen2.5-3B/Syc L18	+0.6	+0.6	–	–	**+0.0**	+0.6	+0.6	–	**+0.0**	**+0.0**	+0.6	+0.6	–	**+0.0**	**+0.0**	+0.6	+0.6	–	**+0.0**	**+0.0**
Mean (pp)	+2.6	+4.3	+7.3	+9.6	+18.3	+1.3	+4.0	+9.3	+8.9	+18.6	+0.6	+4.0	+9.7	+8.9	+18.6	+0.4	+2.7	+9.5	+8.9	+18.6
TanGCE W/T/L			12/21/6					14/22/3					14/23/2					10/26/3		